

Boulder County Health Department Air Inspection Report

COUNTY NUMBER: 013

SOURCE NUMBER: 003

DATE: 06/20/00

COMPANY: Southdown, Inc.

SITE LOCATION: 5134 Ute Highway, Lyons

CONTACT PERSON: Steve Mossberg

TELEPHONE NUMBER: 303-823-2113

INSPECTION TYPE: Routine ☒ Major

Surveillance ☐

TOTAL HOURS REQUIRED: 71.5

Prep/Travel: 8.5

Inspection: 8.0

Report: 55.0

Compliance Assist: ☐ n ☐ y/n

INSPECTOR: Joni Rix

COUNTY: Boulder

TIME: 9:00 am

PERMIT NUMBER: 95OPBO082

Complaint ☐

FA ☐

P2 Assistance: ☐ n ☐ y/n

Emission Points Subject to NSPS/MACT?: Yes - NESHAP, Subpart LLL:Portland Cement on 6/10/2002

This was major source inspection per the 2000 APCD/BCHD contract. The inspection was conducted by myself and Steve Mossberg, Facilities compliance for Southdown. At the time of the original inspection on 6/20, much of the plant was not operating due to high winds. Winds were measured at 45 m.p.h. A follow-up inspection was conducted 6/27 to check on opacity and the current emission readings. When the report refers to actual numbers or opacity, these are referring to the 6/27 follow-up inspection if not indicated.

This facility manufactures Portland Cement. Limestone and other raw materials are extracted from the quarry and are processed through a primary crusher at the Dowe Flats quarry. The crushed material is transported to the plant primary crusher on a 2 mile conveyor belt system and discharged to a stockpile. The stockpiled material is placed on another belt by means of a front end loader or in the future by an underground reclamation system, and goes through the plant primary crusher, processed through a dryer and then a secondary crusher. The material from the secondary crusher is stored in raw material silos. These storage silos contain silica and iron ore and various quarried raw materials. Material from these storage is discharged to weigh belts for the formulation of the desired product. The weigh belts discharge to the raw mill, where the material is mixed and crushed and blended with other raw materials. The homogenized material is delivered to storage silos. From the storage silos the material is delivered to the calciner portion of the kiln. In the kiln the material is formed into small balls known as clinker. Natural gas, coal, petroleum coke, and shredded tires may be used as fuel for the kiln. The clinker is discharged to the cooler, and then moved to internal storage in an A-frame building or outside storage stockpiles. The finished clinker is the raw material for the finish mill. In the finish mill, the clinker is combined with gypsum and ground to a fine material, passed through coolers and stored in the product silos. The material in the product silos can be loaded for bulk transport or sent to a packaging system.

RECEIVED

1. **P017- Dowe Flats Fugitive Dust Sources**
P018 - Lyons Quarry

OCT 18 2000

AIR POLLUTION CONTROL STATIONARY SOURCES PROGRAM			
Parameter	Limitations	Emission Factors	Actual Process/Emissions
Process Rate	3,200,000 tons/year		2,914,903 tons/year

	400,000 tons/month 25,000 tons/day		242,908 tons/month 12,784 tons/day
PM	Dowe Flats Quarry Operations - 134.2 tons/ year	See Condition 1.2	Dowe Flats 114.04 tons/year
	Disturbed Areas @ Lyons Quarry - 19.0 tons/year		Lyons Quarry 11.4 tons/year
PM ₁₀	Dowe Flats Quarry Operations - 58.4 tons/year 7.3 tons/month 916 lbs/day	See Condition 1.2	Dowe Flats 45.66 tons/year 3.8 tons/month .0001 lbs/day
	Disturbed Areas @ Lyons Quarry - 9.4 tons/year		Lyons Quarry 5.64 tons/year

- 1.1 Total material (includes: topsoil, overburden, limestone, and waste rock) handled shall not exceed the limits listed above. Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. (Construction Permit 93BO1414F, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve month total requirement).

Total material has not exceeded the limits listed above. See chart above for throughputs. For 1999, monthly limits were tracked and for this year it is an annual limit.

Monthly and annual records of the amounts of materials handled shall be kept on-site and made available for Division review upon request. The permittee shall maintain records of the number of days of operation during each month. Compliance with the daily limits shall be monitored by dividing the monthly handling rates by the monthly number of days of operation. Records of the number of days of operation and the daily handling rate calculations shall be maintained and made available for inspection upon request.

Monthly and annual records are kept on-site and made available to the Division for review. Records of the days of operation are kept and compliance with the daily limits are calculated by dividing monthly handling rates by the days of operation. These records are available for review.

- 1.2 PM and PM₁₀ emissions shall not exceed the limits listed in the table above. Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. Compliance with the emission limits shall be monitored by not exceeding the process limits set forth in Condition 1.1, and by application of control measures, as set forth in Condition 1.3, below. (Construction Permit 93BO1414F, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve month total requirement).

PM and PM₁₀ emissions have not exceeded the limits listed above. See the table above for actual emissions. Compliance with this condition will be determined by not exceeding the process limits above. In 1999, the process limits were not exceeded. Emissions for Lyons Quarry is based on a disturbed area of 30 acres and emission factors of .380 tons/acre-year

for PM and .187 tons/acre-year for PM₁₀. So far for 2000, source has not exceeded the monthly process limits.

For APEN reporting and fee purposes, the annual PM and PM₁₀ emissions shall be estimated using the actual annual processing rates recorded under Condition 1.1, appropriate emission factors and equations, and control measure efficiencies. A record of the annual emission calculations shall be maintained and made available to the Division for inspection upon request. **For APEN reporting and fee purposes, annual PM and PM₁₀ emissions are based on the actual annual processing rates, emission factors and control measure efficiencies. A record of the annual emission calculations are maintained and available for review.**

- 1.3 Fugitive particulate matter emissions shall be controlled in accordance with the following (Construction Permit 93BO1414, Colorado Regulation No. 1, III.D.1.b).
 - 1.3.1 Mining Activities, including mined land reclamation - Visible emissions shall not exceed 20%, and no off-property transport of visible emissions. **Mining activities in the Dowe Flats area did not exceed 20% and there was no off property transport of emissions.**
 - 1.3.2 Haul Roads - No off-property transport of visible emissions shall apply to on-site haul roads, and the nuisance guidelines shall apply to off-site haul roads. **No off property transport of emissions on haul roads was noted at the time of inspection.**
 - 1.3.3 Haul Trucks - No off-property transport of visible emissions except that when operating off the property of the owner or operator, the applicable guidelines shall be no off-vehicle transport of visible emissions. **No off property transport of emissions from haul vehicles was noted at the time of inspection.**
 - 1.3.4 Adequate soil moisture must be maintained in topsoil and overburden to control emissions during removal. **Adequate soil moisture is maintained by natural moisture and with a water truck. The Dowe Flats quarry is in the process of installing a sprinkler system.**
 - 1.3.5 Topsoil and overburden stockpiles shall be reclaimed and revegetated in accordance with the MLRB permit conditions. Open acreage (mine pits and stockpiles) shall be minimized and in no circumstances shall they be in excess of MLRB or APCD permits, whichever is more restrictive. (Construction Permit 93BO1414F, Attachment A, as modified per Section 1, Condition 1.3 of this permit). **Topsoil and overburden stockpiles are reclaimed and revegetated in accordance with the MLRB permit conditions. Open acreage is limited to 90 acres at any one time.**
 - 1.3.6 Emissions from material handling (i.e. removal, loading and hauling) shall be controlled by watering at all times, except during below-freezing temperatures, unless natural moisture is sufficient to control emissions. A sprinkler system shall be installed and operated to wet muck piles prior to loading, hauling and crushing. Spillage and accumulation of particulate matter shall be cleaned up and shall be managed to insure they do not contribute to fugitive emission during operation. (Construction Permit 93BO1414F, Attachment A, as modified per Section 1, Condition 1.3 of this permit). **Emissions are controlled by watering. A sprinkler system is under construction for this area and for the time being a water truck is used to water. At the time of inspection, the water truck was operating in the Dowe Flat pit.**
 - 1.3.7 Activities causing fugitive dust emission shall be suspended when wind speeds reach or exceed 30 miles per hour, averaged over a 60-minute period. Only those activities affected by wind speed, and for which it is possible to "suspend operation" need to be shut down (i.e. the permittee cannot "shut down" storage piles, thus this condition would not apply to storage piles). Activities may continue when the average wind speed drops

below 30 m.p.h. (Construction Permit 93BO1414F, Attachment A, as modified per Section 1, Condition 1.3 of this permit). **When wind speed reaches 30 m.p.h., the crusher and other fugitive emission activities are suspended. The plant does not do an average 60-minute period for this point, they have agreed to suspend operation as soon as the winds reach 30 m.p.h. At the time of initial inspection, winds were 42 m.p.h. so operation had been suspended.**

- 1.3.8 Vehicle speed on unpaved roads and disturbed areas shall not exceed a maximum of 35 m.p.h. Speed limit signs shall be posted. **Speed limits are not to exceed 25 m.p.h. although at the time of inspection speed limit signs were not posted in the Dowe Flats area. Steve agreed to post one at the front entrance to the mining area.**
- 1.3.9 Unpaved haul roads shall be treated with chemical dust suppressants to maintain a surface crust, and watered, as often as needed to control fugitive particulate emissions. **Unpaved roads are treated with chemical dust suppressants every couple of months in the summer months. This usually starts in April and continues through October. Records of applications of dust suppressant is kept on file for review. Two water trucks also water the plant and records are maintained with the gallons of water used.**
- 1.3.10 Drills shall be equipped with bag collectors to control emissions. **Drills are equipped with bag collectors to control emissions.**
- 1.3.11 Sequential blasting shall be employed. **Sequential blasting is used.**
- 1.3.12 Reclamation work and sequential extraction of material shall be initiated to keep the total disturbed areas at any one time to a minimum. **Reclamation and extraction is done sequentially to keep total disturbed areas to a minimum. Only 90 acres is exposed at any one time.**
- 1.3.13 The permittee shall maintain a copy of the facility's Mining and Reclamation plan (as submitted and approved by the Colorado Department of Minerals and Geology - Mine, Land, and Reclamation Division) on site for Division inspection upon request. (Construction Permit 93BO1414F, Attachment A, as modified per Section 1, Condition 1.3 of this permit). **A copy of the Mining and Reclamation plan is kept on site for review.**

Records of application of dust suppressants shall be maintained on site for inspection upon request. **Records of application of dust suppressants is maintained on site. Beginning in April and ending in September or October, chemical dust suppressants are applied every two months. Records indicate that so far this year applications were made in April, and May.**

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when average wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h., averaged over a sixty minute period. The Annual Certification shall include a statement that the activities identified in Condition 1.3.7 were suspended when average wind speeds reached or exceeded the 30 m.p.h. threshold.

A wind speed instrument (anemometer) is installed on the top of the crusher at Dowe Flats. The unit sends information back to the control room and continuously displays the current wind speed. This information is stored in the computer and print-outs can be made during any time frame to illustrate wind speed. This unit is calibrated annually. At the time of initial inspection (6/20), winds were measuring 42 m.p.h. and consequently much of work at the plant was suspended. At re-inspection on 6/27, winds were 7 m.p.h. per the anemometer.

A weekly inspection of the site shall be conducted to ensure the emission control elements are in place and effective. In addition, at any time when a fugitive dust problem is observed, the permittee shall take action to correct the problem. The permittee shall maintain records of the date and time of any fugitive dust problem observed, and the type and time of action taken to correct the problem. These records shall be maintained on site for inspection upon request. The Annual Certification shall include a statement that the PM Emission Control Plan measures are in place and effectively minimizing fugitive emissions.

A daily inspection of the site is conducted and this was started in February 2000. Fugitive dust problems are noted on the observation sheet with the time and date and a photo. Southdown needs to document better when the problem was fixed and what the correction was. Steve stated they will begin doing this. The inspection sheets are maintained on-site in a binder and available for review. The State Health department did a fugitive dust inspection on April 13th, 2000 and found the plant to be in compliance with their fugitive dust control plan. At the time of my inspection, no fugitive dust was noted, but there were piles of cement dust under the finishing mill and other buildings where the sweeper can't reach that could cause a problem if it got windy.

There shall be no mining of limestone/raw materials or overburden materials at the Lyons Quarry. Reclamation activities (through 1999) and cement kiln dust disposal (as permitted under Section II, Conditions 13 of this permit) are allowed in the Lyons Quarry. Also, there shall be no hauling-in of raw materials from the Larimer Quarry. (Construction Permit 93BO1414F)

No mining is occurring in the Lyons Quarry and reclamation of the area has been completed except for the use of the "C" pit for the disposal of CKD and benification dust. This pit is equipped with a sprinkler system that operates on a timer to control fugitive dust emissions. No raw materials are hauled in from the Larimer Quarry.

The Annual Certification shall include a statement that the facility is in compliance with the restrictions for the Lyons and Larimer Quarries.

Annual certification is due February 1, 2001 and will include a compliance statement.

2. **P017- Dowe Flats Quarry**
S055 - S064
Primary Crusher
Belt Conveyor
Radial Stacker to Stockpiles

Parameter	Permit Limitations	Emission Factors	Actual Process/Emissions
Process Rate	1,050,000 tons/year 400,000 tons/month		858,893 tons/year 71,574 tons/month
PM	<i>Crusher</i> – 0.05 gr/dscm ----- <i>Conveyor</i> – See condition 2.3 ----- 0.14 tons/year	----- <i>Crusher:</i> 0.020 lb/ton <i>Conveyor:</i> .00124 lb/ton (total for all transfer points)	<i>Crusher</i> – 0.05 gr/dscm ----- <i>Conveyor</i> –See 2.3 ----- .1115 tons/year
PM₁₀	0.07 ton/year 0.01 ton/month	<i>Crusher:</i> .009 lb/ton <i>Conveyor:</i> .00059 lb/ton (total for all transfer points)	.0558 tons/year .004 tons/month

- 2.1 Processing and conveying of raw materials at the crusher and conveyor system shall not exceed 1,050,000 tons per year and 400,000 tons per month (Construction Permit 94BO593).

Processing and conveying of raw materials has not exceeded the above limits. See table above for actual throughputs. Numbers are for 1999 throughputs.

Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. Records of the monthly and annual amounts processed shall be maintained and made available to the Division for inspection upon request. (Construction Permit 93BO1414F, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve month total requirement).

For 1999, monthly limits were tracked and for 2000 it is an annual limit. Records for 2000 indicate that they are in compliance with this limit. Records are maintained and available for inspection. See attached 2000 record.

- 2.2 The crusher is subject to the provisions of 40 CFR Part 60, Subpart OOO, Standards of Performance for Non-Metallic Mineral Processing Plants, as adopted by reference in Colorado Regulation No. 6, Part A, as set forth in Condition 20 of this permit, except that stack testing is not required during the initial five year term of this permit.

The crusher is subject to the provisions of 40 CFR Part 60, Subpart OOO, see Condition 20 of this permit for specifics. Stack testing was done on the crusher earlier this year. Limits and compliance with this condition are listed above in the table - specifically, the PM and opacity standards. See Condition 20 of permit for specifics.

- 2.3 During any consecutive sixty (60) minute period, PM emissions from the conveyor operation shall not exceed:

$$E = 17.31 (P)0.16$$

Where:

E is the allowable particulate emissions in lbs/hr.

P is the process weight rate in tons/hr

(Colorado Regulation No. 1, III.C.1.b and Colorado Regulation No. 6, Part B, II.C.2). In absence of evidence to the contrary, compliance with this limit shall be monitored using the production records required in Condition 2.1 and the baghouse operating and maintenance requirements of Condition 20.

Compliance with this limit is monitored using the production records and the baghouse operating and maintenance requirements.

- 2.4 Annual emissions of PM and PM₁₀ shall not exceed the limits listed in the above table (Construction Permit 94BO593, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve month total requirement). Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. The throughput records required in Condition 2.1 and the bagfilter operating and maintenance procedures required in Condition 2.2.2 shall be used to monitor compliance with the emissions limits.

Annual emissions of particulates shall not exceed the limits in the table above. See table above for actual emissions calculated with 1999 throughputs. Compliance for 1999 was

shown monthly and for 2000 will be an annual amount. The throughput records and bagfilter operating and maintenance procedures are used to monitor compliance.

For APEN reporting purposes, annual estimated PM and PM₁₀ emissions (tons/year) shall be calculated using the actual annual amounts processed, the emission factors listed in the table above (Emission factors taken from Construction Permit 94BO593 analysis), and appropriate control equipment collection efficiencies. Records of the emissions estimates shall be maintained and made available to the Division for inspection upon request.

NOTE: If the bagfilters are maintained and inspected properly, as required under Condition 2.2.2, the following control efficiencies may be assumed for emission calculations: 98.6% for enclosed conveyor and baghouses and for the crusher baghouse.

PM and PM₁₀ emissions (tons/year) are calculated using actual amounts processed, emission factors listed in the table and control equipment efficiencies.

- 2.5 The conveyor system is subject to the provisions of 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants, as adopted by reference in Colorado Regulation No. 6, Part A, as set forth in Condition 21 of this permit.

The conveyor is subject to the 40CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants. See Condition 21 of this permit for the specifics.

- 2.6 These sources are subject to the opacity limits set forth in Condition 22 of this permit.
Opacity is checked daily with a pre-printed check list. If person sees emissions, the baghouse is checked to see if there is a problem. Each month a specific work order for the baghouses is printed out and the bags are inspected. Bags are changed when a problem is noted. See Condition 22 of this permit for specifics on opacity.

**3. P004 - Storage and Handling of Raw Materials at Plant Site
S009 - Underground Reclamation System
Front End Loader Activity**

Parameter	Limitations	Emission Factors	Actual Process/Emissions
Process Rate	<i>Underground Reclamation System:</i> 1,050,000 tons/year 120,750 tons/month 4,000 tons/day <i>Front End Loaders:</i> 193,200 tons/year 25,000 tons/month 1,300 tons/day		<i>Underground Reclamation System:</i> Not installed yet. New permit received in August 2000 allows: 966,000 tons per year 120,750 tons/month 4,000 tons/day **See 3.1**
PM	2.5 tons/year	Front End Loader: 0.26 lb PM/ton .0114 lb PM ₁₀ /ton Underground Reclamation: .004 lb PM/ton .002 lb PM ₁₀ /ton	To date not tracked since underground reclamation system has not been installed.

PM ₁₀	1.1 tons/year 0.14 ton/month 15 lbs/day	same as above	Same as above
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- 3.1 Process rates shall not exceed the rates listed in the table above (Construction Permit 98BO0292, revised according to Section I, Condition 1.3 of this permit, to correct the Underground Reclamation System throughput from 966,000 to 1,050, 000 tons/year, and to delete the rolling twelve month total requirement). Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. The permittee shall maintain monthly an annual records of actual process rates on site for inspection upon request. The permittee shall maintain records of the number of days of operation for each month. Compliance with the daily limits shall be monitored by dividing the monthly production rate by the number of days of operation for that month. Records of the number of days of operation and the daily production rate calculations shall be maintained and made available for inspection upon request.

Process rates shall not exceed the rates listed in the table above. The Underground Reclamation System has not been installed yet so none of the above throughputs have been tracked. Southdown has not tracked the front loader activity either since they thought it was part of the underground system. The limitations above may change. Southdown has requested to decrease the PM at the finish mill and increase the emissions at this point so the overall particulate emissions would remain the same. They are waiting to hear from APCD.

- 3.2 PM and PM₁₀ emissions shall not exceed the rates listed in the table above (Construction Permit 98BO0292, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve moth total requirement). Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. The actual material processing records required in Condition 3.1, the PM emission control plan required in Condition 3.6, and the bagfilter operating and maintenance procedures required in this Condition 3.2 shall be used to monitor compliance with the emissions limits. The Annual Certification shall describe how the inspection, maintenance work, and repair documentation affected emissions.

PM and PM₁₀ emissions shall not exceed the rates listed in the table above. The limits have not been tracked since the Underground Reclamation System has yet to be installed. Also see condition above as limitations may change.

For APEN reporting and fee purposed, the annual PM and PM₁₀ emissions shall be estimated using the actual annual processing rates recorded under Condition 3.1, the listed emission factors (emission factors taken from Construction Permit 98BO0292 analysis), and control measure efficiencies. A record of the annual emission calculations shall be maintained and made available to the Division for inspection upon request.

Emissions shall be estimated using the actual processing rates.

NOTE: If the bagfilters are maintained and inspected properly, as required under this condition, the following control efficiencies may be assumed for emission calculations: 99.0% for the underground reclamation system baghouse.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and made available to the Division upon request.

A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request.

- 3.3 The equipment/activities in this process system are subject to 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants, as adopted by reference in Colorado Regulation No. 6, Part A, as set forth in Condition 21 of this permit.
The equipment/activities in this process system are subject to 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants. See Condition 21 of this permit for specifics.
- 3.4 These sources, except for Front End Loader Activity, are subject to the opacity limits set forth in Condition 22 of this permit.
These sources, except for Front End Loader activity, are subject to the opacity conditions in Condition 22 of this permit which is 20%. Compliance is determined with a daily plant walk-through.
- 3.5 The following particulate emission control measures shall be applied to the particulate emission producing sources (Construction Permit 98BO0292, Colorado Regulation No. 1, III.D.1.b).
- 3.5.1 Height of discharge from the stacker belt shall be adjusted to minimize the drop height to less than 3 feet. Water spray bars shall be used if they natural surface moisture is insufficient to limit opacity to less than 10 percent. **Height of discharge from the stacker is adjusted to be less than 3 feet. A three foot measure hangs from the bottom of the stacker that indicates the three foot limit. Water spray bars are installed on the top of the stacker and if they are not operating, the stacker does not operate. In addition, there is an almost three foot long wrap that goes around the drop-off area of the stacker. This has reduced fugitive emissions from the stacker a great deal.**
- 3.5.2 Transfer points shall be enclosed and maintained under negative pressure. **This condition applies to the underground reclamation system that has not been installed.**
- 3.5.3 Moisture content of the materials handled by front-end loaders shall be adequate to effectively control the emissions. (Construction Permit 98BO0292, as modified according to Section I, Condition 1.3 of this permit). **Moisture content of the materials handled by the front-end loaders is maintained to control emissions.**
- 3.5.4 When feeding the primary crusher a the plant, material drop height from the front-end loaders shall be minimized, and the drop height shall not exceed 3 feet above the top edge of the hopper. (Construction Permit 98BO0292, as modified according to Section I, Condition 1.3 of this permit). **Drop height to the primary crusher at the plant does not exceed three feet above the hopper. A line will be marked on the hopper so the front end loaders can see where the three feet is.**
- 3.5.5 The stockpile work area on which the front-end loaders operate shall be treated with chemical dust suppressants and/or water to minimize the generation of fugitive emissions. The operation of front-end loaders in this area will be suspended whenever

the average wind speed exceeds 30 m.p.h. (Construction Permit 98BO0292, as modified according to Section I, Condition 1.3 of this permit). **The stockpile work area is treated with water and with a chemical dust suppressant that is like a foam.**

Operations are suspended when the wind speeds reach 30 m.p.h.

- 3.5.6 Paved travel areas used by the front-end loader shall be regularly swept with a high efficiency vacuum sweeper to minimize material buildups. In addition, these areas will be watered as necessary and vehicle traffic suspended or rerouted if fugitive emissions become a concern. (Construction Permit 98BO0292, as modified according to Section I, Condition 1.3 of this permit). **Paved areas are swept at least once per day with a high efficiency vacuum sweeper. Southdown has two sweepers - the large one does the plant and the smaller one does the tighter spaces.**
- 3.5.7 Front end loader and hauling activities shall be suspended when the wind speed reaches or exceeds 30 m.p.h., averaged over a 60-minute period. Activities may continue when the average wind speed drops below 30 m.p.h. (Construction Permit 98BO0292, as modified per Section 1, Condition 1.3 of this permit). **Operations are suspended when the wind speeds reach 30 m.p.h. averaged over a 60 minute period. Plant uses the anemometer installed on the crusher at the Dowe Flats quarry to know the wind speed.**

The permittee shall install a marker to indicate a distance 3 feet above the top of the hopper. **A marker has not been installed on the hopper to indicate the three feet. Steve stated they will do this.**

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h., averaged over a sixty minute period. The Annual Certification shall include a statement that the activities identified in Condition 3.5.7 were suspended when average wind speeds reached or exceeded 30 m.p.h. threshold.

A wind speed instrument (anemometer) is installed on the top of the crusher at Dowe Flats. The unit sends information back to the control room and continuously displays the current wind speed. This information is stored in the computer and print-outs can be made during any time frame to illustrate wind speed. This unit is calibrated annually. The Annual Certification shall include a statement that the activities identified in Condition 3.5.7 were suspended when average wind speeds reached or exceeded 30 m.p.h. threshold. At the initial inspection on 6/20 the plant was shut down due to winds that were registering 42 miles per hour on the anemometer. A revisit was made on 6/27 and the wind speed was 7 miles per hour per the anemometer.

A weekly inspection of the site shall be conducted to ensure the emission control elements are in place and effective. In addition, at any time when a fugitive dust problem is observed, the permittee shall take action to correct the problem. The permittee shall maintain records of the date and time of any fugitive dust problem observed, and the type and time of action taken to correct the problem. These records shall be maintained on site for inspection upon request. The Annual Certification shall include a statement that the PM Emission Control Plan measures are in place and effectively minimizing fugitive emissions.

A daily walk through of the plant is conducted to check for fugitive emissions. A weekly inspection of the site is conducted. Fugitive dust problems are noted on the observation sheet with the time and date and a photo. Southdown needs to document better when the

problem was fixed and what the correction was. Steve stated they will begin doing this. The inspection sheets are maintained on-site in a binder and available for review.

- 3.6 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT standard, as set forth in Conditions 24 of this permit.
This source will be subject to the MACT on June 10, 2002.

4. **P001**
S002 - Primary Crusher at Plant Site
S004 - Surge Silo

Parameter	Permit Limits	Emission Factors	Actual Process/Emissions
Material Throughput	No Limits - although required to track		886,754 tons/year
PM & PM₁₀	No Limits - although required to track	PM & PM ₁₀ : Crusher: 0.001 lb/ton Surge Silo: 2.9×10^{-5} lb/ton	Crusher: 886.77 lbs/year Surge Silo: 25.7 lbs/year
PM	See Condition 4.3		See 4.3

- 4.1 Records of the annual amounts of raw materials throughput shall be maintained and made available to the Division for inspection upon request.
Records of the annual amounts of raw material throughputs are maintained and available for review. See above table for 1999 throughputs.

- 4.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 4.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). These emission factors account for bagfilter control. Records of the calculations shall be maintained and made available for inspection upon request.
Emissions are estimated using the actual annual raw material throughputs and the emission factors list above. See the above table for actual emissions.

- 4.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed:
 $E = 17.31 (P)0.16$

Where:

E is the allowable particulate emissions in lbs/hr.

P is the process weight rate in tons/hr

(Colorado Regulation No. 1, III.C.1.b). In absence of evidence to the contrary, compliance with this limit shall be monitored using the baghouse operating and maintenance requirements of Condition 22.

Compliance with this limit is monitored using the production records and the baghouse operating and maintenance requirements.

- 4.4 These sources are subject to the opacity limits set forth in Condition 22 of this permit. **These sources are subject to the opacity limits in Condition 22 of this permit. A daily walk through of the plant is conducted to check for opacity problems. At the time of inspection, no opacity was noted.**

5. P002 - Raw Materials Drying (S005)

Parameter	Permit Limits	Emission Factors	Actual 1999 Process/Emissions
Operating Hours	7,000 hours/year		6398.5 hours/year
Dryer Feed	1,050,000 tons/year 160 tons/hour		887,024 tons/year 138.62 tons/hour
Natural Gas	209.32 million scf/year		206.97 million scf/yr
Coal	1.4 tons/hour		Not used in 1999
Dryer Heat Input	210,000 million Btu/year		223,526 mmBtu/year (see 5.3)
PM	22.8 tons/year	Stack Test	22.18 tons/year
PM ₁₀	22.8 tons per year and 6.5 lbs/hour	Stack Test	22.18 tons/year 6.9 lbs/hour (see Note in 5.4)
SO ₂	36.7 tons/year	Stack Test	31.05 tons/year
NO _x	13.9 tons/year	Stack Test	13.31 tons/year
VOC	144.8 tons/year	Stack Test	141.92 tons/year
CO	57.3 tons/year	Stack Test	55.44 tons/year
Lead	1.6 tons/year 1.5 micrograms per standard cubic meter averaged over a one-month period (State-Only)	Stack Test ----- -----	0.44 tons/year (based on emission factor) Stack test was non- detectable ----- Modeling submitted by source to show compliance with the 1.5 mg/cm

- 5.1 Annual operating hours , annual and hourly feed rates, and annual fuel consumption shall not exceed the limits listed in the above table (Construction Permit 12BO444-1). Records of the monthly and annual operating hours, annual and monthly feed rates, and annual fuel consumption shall be maintained and made available to the Division for inspection upon request. Compliance with the hourly feed rate shall be determined by dividing the monthly feed rate by the monthly hours of operation.

Annual operating hours , annual and hourly feed rates, and annual fuel consumption shall not exceed the limits listed in the above table. Records are maintained and available for review. Compliance with the hourly feed rate is determined by dividing the monthly feed rate by the monthly hours of operation. See above for the actual amounts.

- 5.2 The terms and conditions of this permit are based on the dryer using natural gas as the primary fuel. Coal may be used as a backup fuel during emergencies and natural gas curtailments. The Division shall be notified, in writing, within seven (7) calendar days of the start of coal use. Records of the amounts of coal burned and the duration of the combustion must be maintained. Compliance with the hourly coal consumption limit shall be determined by dividing the monthly amount of coal burned by the number of hours coal was burned during the month. (Construction Permit 12BO444-1)

Terms and conditions of this permit are based on using natural gas as the primary fuel. Coal may be used as a backup fuel, but to date coal has not been used for the dryer.

- 5.3 Dryer heat input shall not exceed 0.21×10^6 mmBtu per year (Construction Permit 12BO444-1). Records of the annual amount shall be maintained and made available to the Division for inspection upon request. The heat content of the natural gas shall be determined semi-annually using ASTM Methods or other appropriate methods approved by the Division.

Dryer heat input shall not exceed 0.21×10^6 mmBtu per year (or as listed above - 210,000 mmBtu/year). Records of the annual amounts are maintained, see above table for annual amounts. For 1999, the mmBtu was over the permit limit and letter explaining the reason was sent to the Division (see attached copy). The heat content is determined semi-annually. For 1999, the mmBtu was 1.08 and for 2000 the mmBtu is 1.05. Southdown had to make a correction to their emissions since initially they were using a higher mmBtu to figure out their emissions, which was related to an inaccurate gas meter. This problem was corrected in early April 2000 so this should not be a problem for the 2000 calculations.

If coal is used as a fuel during the semi-annual period, coal shall be sampled at least semiannually to determine the heat content and weight percent sulfur. Prior to the first coal sampling event, a coal sampling plan shall be submitted to the Division for approval. The coal sampling plan shall at a minimum include the elements identified in Appendix G. The sulfur and heat content of the coal shall be determined by sampling and testing the coal in accordance with the Division approved coal sampling plan. Alternatively, vendor receipts, invoices, contracts or similar information may be used to provide the values for sulfur and heat content. Such information shall indicate that the coal sampling and analysis has been performed in accordance with ASTM procedures, or equivalent, identified in Appendix G.

Coal has not been used as a fuel and Southdown understands that semi-annual testing to determine heat content and percent sulfur would be required.

- 5.4 PM and PM₁₀ emissions shall not exceed the limits listed in the table above (Construction Permit 12BO444- 1). Compliance with the emission limits shall be monitored using the records of operating hours, fuel use, and feed rate required in Condition 5.1, the results of the test required in this Condition 5.4 (test results during coal burning shall be used to estimate emissions when coal is burned, and test results during natural gas shall be used to estimate emissions when natural gas is burned), and the baghouse maintenance and operating procedures required in this Condition 5.4. Compliance with the short term PM₁₀ emission limit shall be determined by dividing the monthly fuel consumption by the monthly hours of operation, and applying the stack test results.

PM and PM₁₀ emissions shall not exceed the limits in the table above. Compliance is monitored with the operating hours, fuel use, feed rate, and the baghouse maintenance and operating procedures. Compliance with the short term PM₁₀ emission limit is determined by dividing the monthly fuel consumption by the monthly hours of operation, and applying

the stack test results. See table above for the actual emissions based on throughputs and emission factors.

Note: For 1999, the lbs/hour emission limit was exceeded since they did not operate 7,000 hours, although the overall tons per year is under the permit limit. This number is based on throughput and number of hours of operation and may be overestimated. The first stack test done in early 2000 on the dryer unit failed, and after replacing over 40 bags in the baghouse, the stack test was redone and results were all under permit limits. The stack test showed 8.2 tons/year of particulate emissions and 2.3 lbs/hour particulate emissions – both well under the permit limits.

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 23 of this permit, except that: (1) if the first annual test results indicate emissions are less than or equal to 50% of the emission limit, no additional tests are required during the permit term; (2) if the first annual test results indicate emissions are more than 50%, but less than or equal to 75% of the emission limit, another test is not required until within one year of expiration of this permit; (3) if the first annual test results indicate emissions are greater than 75% of the emission limit, an annual test is required until the provisions of (1) or (2) are met. Tests shall be performed using the primary fuel, natural gas, however, a stack test shall be performed according to Condition 23 of this permit, to determine compliance when coal is used for a fuel, if coal is used for 45 days or more during any calendar year.

Stack testing for PM was done in January (see note above) and again June 9, 2000. Source was in compliance with the permit limits on the second stack test. The June results of the stack test are currently being reviewed by the Division. See attached copy of test results.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouse shall be conducted in accordance with manufacturer's specifications and/or the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request. For APEN reporting and fee purposes, annual PM and PM10 emissions shall be calculated using the test results and the actual operating hours as recorded under Condition 3.1. Records of the emissions calculations shall be maintained and made available for inspection upon request.

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of emissions calculations are maintained and calculated using the test results and the operating hours.

- 5.5 Emissions shall not exceed the limits listed in the table above (Construction Permit 12BO444-1, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve month total requirement). Compliance with the emission limits shall be monitored using the monthly records of operating hours, fuel use, and feed rate required in Condition 5.1, the results of the test required in this Condition 5.5 (test results obtained during coal burning conditions shall be used to estimate emissions from burning coal, and test results obtained during natural gas burning shall be used to estimate emissions when natural gas is used), and monitoring operating parameters, as required in this Condition 5.5.

Emissions shall not exceed the limits listed in the table above. Compliance with the emission limits are monitored using the monthly records of operating hours, fuel use, and feed rate required in Condition 5.1. See table above for actual emissions.

Stack Testing

Performance testing for lead, sulfur dioxide, nitrogen oxides, carbon monoxide, and volatile organic compounds shall be performed within one year of issuance of this permit, and every five years thereafter, in accordance with the requirements and procedures set forth in the following EPA Test Methods:

Method 12 (lead)
Method 6-6C (SO₂)
Method 10 (CO)
Method 7E (NO_x)
Method 25A (VOC)

Tests shall be performed under natural gas conditions, however, tests shall be performed under coal burning conditions if a test as described under Condition 5.4 is required (a coal test is required only once per permit term, if applicable). A stack testing protocol shall be submitted for Division approval at least thirty (30) calendar days prior to performance of each test. No stack test shall be performed without prior written approval by the Division. The Division reserves the right to witness the test. The required number of copies of the compliance test results shall be submitted to the Division within forty-five (45) calendar days of the completion of the test.

Stack testing was done in January and again June 9, 2000. Results indicate that the source is in compliance with the permit limits and the results are currently under review with the Division. See attached results of particulate and lead stack testing done June 9, 2000.

Operating Parameters

Operating parameters shall be established during each performance test. The permittee shall operate the dryer within the established parameters until new parameters are established during the next test. The test protocol shall identify and describe which parameters will be measured and how the parameters are related to emissions.

Operating parameters were established prior to the performance test. These parameters include sample volume, moisture, gas velocity, gas flow, temperature, etc.

Within six months of completion of the first required test, the permittee shall submit to the Division a plan for monitoring the parameters identified during the test.

The plan for monitoring will include the broken bag indicator and will be submitted. Steve will check with the state to see if this is adequate monitoring or if more is needed.

For APEN reporting and fee purposes, annual emissions shall be calculated using the test results and the actual annual operating hours as recorded under Condition 5.1. Records of the calculations shall be maintained and made available for inspection upon request.

Annual emissions are calculated using test results and the actual operating hours. Records are maintained and available for inspection.

- 5.6 Emissions of lead shall not be such that emissions result in an ambient lead concentration exceeding 1.5 micrograms per standard cubic meter averaged over a one-month period (Colorado Regulation No. 8, Part C, Section I.B). The permittee submitted modeling results which indicate compliance with this applicable requirement. A copy of the modeling analysis and results shall be maintained and made available for inspection upon request.

Emissions of lead do not exceed 1.5 micrograms per standard cubic meter averaged over a one-month period. The stack test results of the lead testing are listed above and the permittee submitted modeling results which indicate compliance with this applicable requirement.

- 5.7 This source is subject to the requirements of 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants as adopted by reference in Colorado Regulation No. 6, Part A, as set forth in Condition 21 of this permit. Note: On and after the MACT compliance date of June 10, 2002, any affected source subject to the provisions of the MACT standard is exempted from any otherwise applicable new source performance standard contained in 40 CFR part 60, Subpart F. (63.1356)

This source is subject to the requirements of 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants. Once the MACT is in effect (June 10, 2002), this source will be exempt from the new source standards within 40 CFR Part 60.

- 5.8 These sources are subject to the opacity limits set forth in Condition 22 of this permit.
This source is subject to the opacity limits in Condition 22 (20%). At the time of the inspection, no opacity was noted.

- 5.9 This source shall utilize Reasonably Available Control Technology (RACT) for VOC emissions (Colorado Regulation No. 7, II.C). Operation of this dryer as designed represents RACT. Any modification of the design shall require a new RACT determination and modification or reopening of this permit.

Operation of the dryer as designed is considered to be RACT for VOC emissions. Any modification will require a new RACT determination.

- 5.10 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT standard, as set forth in Conditions 24 of this permit.
This source will be subject to the MACT on June 10, 2002.

**6. P003 - Secondary Crushing
Crusher and Screen
S001 - Secondary Crushing Dust Silo
S003 - #4 & #6 Belt Transfer**

Parameter	Permit Limit	Emission Factors	Actual Process/Emissions
Process Rate	No Limit - but required to track		858,893 tons/year
PM & PM ₁₀	No Limit - but required to track	PM & PM ₁₀ : Screening and Crushing: 0.31 lb/ton Silo and Belt Transfer: 2.9 x 10 ⁻⁵ lb/ton (each bagfilter)	Screening and Crushing: 266,256 lbs/year Silo and Belt Transfer: 24.9 lb/ton (each bagfilter)
PM	See Condition 6.3		See 6.3

- 6.1 Records of the annual amount of raw material throughput shall be maintained and made available to the Division for inspection upon request.

Records are maintained with an annual amount of raw material throughputs and available for review. Throughputs are actually the same as the primary crusher at Dowe Flats. See above table for actual throughputs.

- 6.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 6.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). These factors take the control equipment efficiency into account. Records of the calculations shall be maintained and made available for inspection upon request.

Annual emissions are based on actual annual raw material throughputs. Records of emissions calculations are maintained and calculated using the emission factors above. See table above for actual emissions.

- 6.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed:

$$E = 17.31 (P)^{0.16}$$

Where:

E is the allowable particulate emissions in lbs/hr.

P is the process weight rate in tons/hr

(Colorado Regulation No. 1, III.C.1.b). In absence of evidence to the contrary, compliance with this limit shall be monitored using the baghouse operating and maintenance requirements of Condition 22.

Compliance with this limit is monitored using the baghouse operating and maintenance requirements as per Condition 22 of this permit. Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request.

- 6.4 These sources are subject to the opacity limits set forth in Condition 22 of this permit. **These sources are subject to the opacity in Condition 22 of this permit, which for this point is 20%. At the time of inspection, no opacity was seen. A daily walkthrough of the plant is conducted and documented to identify any opacity problems.**

**7. P004 - Raw Materials Storage
S006 through S008 - Raw Materials Storage Silos**

Parameter	Permit Limit	Emission Factors	Actual Process/Emissions
Process Rate	No Limit Specified		845,049 tons/year
PM & PM ₁₀	No Limit Specified	PM & PM ₁₀ 2.9×10^{-5} lb/ton (for each baghouse stack)	.01225 lbs/year (for each baghouse)

PM	See Condition 7.3		See 7.3
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- 7.1 Records of the annual amounts of raw material throughputs shall be maintained and made available to the Division for inspection upon request.
Records are maintained with an annual amount of raw material throughputs and available for review. Records are updated monthly and emissions are calculated. See attached copy. Plant has six silos – 2 with 1st ridge, 2 with 3rd ridge, 1 for Type 1 cement, 1 for 2nd ridge and weather rock. These silos feed the raw mill and are controlled with three dust collectors referred to S006, S007 and S008. See above table for actual throughputs for 1999.
- 7.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 7.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). This factor accounts for the control equipment efficiency. Records of the calculations shall be maintained and made available for inspection upon request.
Annual emissions are based on actual annual raw material throughputs. Records of emissions calculations are maintained and calculated using the throughputs and the emission factor listed above. See above for actual emissions.
- 7.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed:

$$E = 17.31 (P)0.16$$
Where:
E is the allowable particulate emissions in lbs/hr.
P is the process weight rate in tons/hr
- (Colorado Regulation No. 1, III.C.1.b). In absence of evidence to the contrary, compliance with this limit shall be monitored using the baghouse operating and maintenance requirements of Condition 22.
Compliance with this limit is monitored using the baghouse operating and maintenance requirements as per Condition 22 of this permit. Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request.
- 7.4 These sources are subject to the opacity limits set forth in Condition 22. of this permit.
These sources are subject to the opacity in Condition 22 of this permit, which for this point is 20%. At the time of inspection, no opacity was seen. A daily walkthrough of the plant is conducted to identify any opacity problems.
- 7.5 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT requirements, as set forth in Conditions 24 of this permit.
This source will be subject to the MACT on June 10, 2002.
8. **P005 - Raw Mill Grinding
S010 - Raw Mill
S011 - Raw Mill Auxiliary (Separator)
S012 - Raw Mill Weigh Feeder**

S013 - Iron/Silica Silo

Parameter	Permit Limit	Emission Factor	Actual Process/Emissions
Process Rate	No Limits Specified		845,049 tons/year
PM & PM ₁₀	No Limits Specified	PM & PM ₁₀ : S010 - 0.012 lb/ton S011 - 0.032 lb/ton S012 - 0.019 lb/ton S013 - 0.0031lb/ton	S010 – 10,140 lbs/year S011 – 27,041 lbs/year S012 – 16,055 lbs/year S013 – 2,619 lbs/year
PM	See Condition 8.3		See 8.3

- 8.1 Records of the annual amounts of raw material throughputs shall be maintained and made available to the Division for inspection upon request.
Records are maintained with an annual amount of raw material throughputs and available for review. See above table for actual throughputs in 1999.
- 8.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 8.1, and the emission factors listed in the table above (AP-42, Section 11.6, January, 1995). These factors account for the control equipment efficiency. Records of the calculations shall be maintained and made available for inspection upon request.
Annual emissions are based on actual annual raw material throughputs. Records of emission calculations are maintained and calculated using the annual throughputs and the emission factor above. See above for actual emissions.
- 8.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed:

$$E = 17.31 (P)^{0.16}$$
Where:
E is the allowable particulate emissions in lbs/hr.
P is the process weight rate in tons/hr
- (Colorado Regulation No. 1, III.C.1.b). In absence of evidence to the contrary, compliance with this limit shall be monitored using the baghouse operating and maintenance requirements of Condition 22.
Compliance with this limit is monitored using the baghouse operating and maintenance requirements as per Condition 22 of this permit. Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request.
- 8.4 These sources are subject to the opacity limits set forth in Condition 22 of this permit.
These sources are subject to the opacity in Condition 22 of this permit, which for this point is 20%. At the time of inspection, no opacity was seen. A daily walkthrough of the plant is conducted to identify any opacity problems.

- 8.5 On and after the MACT compliance date of June 10, 2002, the raw mill is subject to the MACT standard, as set forth in Conditions 24 of this permit.

This source will be subject to the MACT on June 10, 2002.

9. **P006 - Homogenizing and Blending**
S014 - Homogenizing Silo
S015 - Kiln Feed Silo

Parameter	Permit Limitations	Emission Factors	Actual Process/ Emissions
Process Rate	No Limitations		845,049 tons/year
PM & PM ₁₀	No Limitations	PM & PM ₁₀ : 2.9 x 10 ⁻⁵ lb/ton (for each baghouse stack)	24.5 lbs/year (for each baghouse stack)
PM	See Condition 9.3		See 9.3

- 9.1 Records of the annual amounts of material throughput shall be maintained and made available to the Division for inspection upon request.

Records are maintained with an annual amount of raw material throughputs and available for review. See above table for actual throughputs for 1999.

- 9.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual material throughputs recorded under Condition 9.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). This factor accounts for control equipment efficiency. Records of the calculations shall be maintained and made available for inspection upon request.

Annual emissions are based on actual annual raw material throughputs. Records of emissions calculations are maintained and calculated using the annual throughputs and emission factors listed above. See above for actual emissions calculated from the 1999 throughputs.

- 9.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed:

$$E = 17.31 (P)0.16$$

Where:

E is the allowable particulate emissions in lbs/hr.

P is the process weight rate in tons/hr

(Colorado Regulation No. 1, III.C.1.b). In absence of evidence to the contrary, compliance with this limit shall be monitored using the baghouse operating and maintenance requirements of Condition 22.

Compliance with this limit is monitored using the baghouse operating and maintenance requirements as per Condition 22 of this permit. Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request.

- 9.4 These sources are subject to the opacity limits set forth in Condition 22 of this permit.

These sources are subject to the opacity in Condition 22 of this permit, which for this point is 20%. At the time of inspection, no opacity was seen. A daily walkthrough of the plant is conducted to identify any opacity problems.

- 9.5 On and after the MACT compliance date of June 10, 2002, the raw mill is subject to the MACT standard, as set forth in Condition 24 of this permit.

This source will be subject to the MACT on June 10, 2002.

10. **P007- Kiln, Calciner (S016)**
P008 - Clinker Cooler (S018)

Parameter	Permit Limitations	Emission Factors	Actual Process/ Emissions
Kiln Feed Rate & Clinker Production Rate		Feed = 845,049 tpy Clinker = 500,031 tpy	1,345,080 tons/year (combined total of kiln feed and clinker production for 1999)
Operating Hours	8,064 hours/year		7,953 hours/year
Kiln Feed Rate	120 tons/hour 967,680 tons/year (dry basis)		106.26 tons/hour 845,049 tons/year
Natural Gas	The use of any other fuel requires Division approval		Not used in 1999
Coal	" "		12 tons/hour
Whole or Shredded Tires	" "		Not used in 1999
Petroleum Coke	" "		Not used in 1999
PM & PM ₁₀	133 tons/year	Stack Test	126.8 tons/year
PM & PM ₁₀ - Kiln	0.30 pound/ton (dry basis) ----- 0.275 pound/ton of feed (dry basis)	Stack Test	.023 lb/ton of feed
PM & PM ₁₀ - Clinker Cooler	0.10 pound/ton of feed (dry basis)	Stack Test	.025 lb/ton of feed
NO _x	2649 tons/year		1973.7 tons/year
CO	396 tons/year		135.1 tons/year
SO ₂	1340 tons/year 7 lbs/ton of material		14.0 tons/year 0.09 lbs/ton
VOC	138 tons/year		2.8 tons/year
HCl		0.0381 lb/ton clinker	19,051 lbs/year
Lead	4.4 tons/year ----- 1.5 micrograms per standard cubic meter averaged over a one-	Stack Test -----	0.473 tons/year ----- Modeling submitted to APCD, and seem to be in compliance with this

	month period (State- Only)		condition – See 10.12
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- 10.1 The permittee shall record the daily production rates and kiln feed rates (40 CFR Part 60, Subpart F, as adopted by reference in Colorado Regulation No. 6, Part A and Construction Permit 12BO444-2). In addition, the permittee shall maintain a record of daily raw material consumption and clinker production rates (Construction Permit 12BO444-2). Note: After the MACT compliance date of June 10, 2002, this requirement still applies, as a Construction Permit applicable requirement.

Records are maintained with the daily production rates and kiln feed rates, which is the same as what goes into the raw mill. The daily raw material consumption and clinker production rates are made into monthly amounts for calculating emissions. See above table for actual throughputs.

- 10.2 Annual operating hours shall not exceed 8,064 (Construction Permit 12BO444-2). The permittee shall maintain records of annual and daily (Construction Permit 12BO444-2) operating hours on site for inspection upon request.

Operating hours are recorded in the monthly production reports. See above table for actual hours.

- 10.3 The Construction Permit was issued based on permitted fuels consisting of natural gas, coal, and/or used tires. The use of petroleum coke is incorporated directly into this operating permit according to Section 1, Condition 1.3 of this permit. No other fuels shall be used without prior approval from the Division. Records of the amount of each type of fuel shall be maintained on an annual and daily (Construction Permit 12BO444-2) basis and made available for inspection upon request. Kiln feed rate shall not exceed 120 tons/hour or 967,680 tons/year (dry basis). (Feed rate limits incorporated directly into this operating permit.) Records of the annual kiln feed rate shall be maintained and made available for inspection upon request. Compliance with the hourly kiln feed rate shall be determined by dividing the daily kiln feed rate, as recorded under Condition 10.1, by the daily hours of operation, as recorded under Condition 10.2.

Coal is the major fuel used along with some natural gas. At this time no other fuels are used and permittee understands that approval from the division is required if alternate fuels are used. Records are maintained on the amount of fuel and the kiln feed rate. See above table for actual throughputs.

- 10.4 This source is subject to the requirements of 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants, as adopted by reference in Colorado Regulation No. 6, Part A, as follows.

10.4.1 Particulate matter emissions from the kiln shall not exceed 0.30 pound per ton of feed (dry basis). **See table for particulate emissions.**

10.4.2 Particulate matter emissions from the clinker cooler shall not exceed 0.10 pound per ton of feed (dry basis). **See table for particulate emissions.**

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 23 of this permit, except that: (1) if the first annual test results indicate emissions are less than or equal to 50% of the emission limit, no additional tests are required during the permit term; (2) if the first annual test results indicate emissions are more than 50%, but less than or equal to 75% of the emission limit,

another test is not required until within one year of expiration of this permit; (3) if the first annual test results indicate emissions are greater than 75% of the emission limit, an annual test is required until the provisions of (1) or (2) are met. Tests shall be performed using the primary fuel that will be used during the year or the permit term, whichever is applicable, however, a stack test shall be performed according to Condition 23 of this permit, to determine compliance when any other fuel is used, if any other fuel is used for 45 days or more during any calendar year. Alternatively, the permittee may test using the worst case particulate matter emitting fuel, as approved by the Division, and shall then use this emission rate to estimate PM emissions from all fuels. The emission rate of particulate matter shall be computed for each run using the procedures and equation set forth in 40 CFR Part 60, Subpart F, 60.64.

Stack testing was completed in February and April for this point by APT. Emissions calculated based on 8,064 hours of operation. See attached copy.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented, maintained and made available to the Division upon request.

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of emissions calculations are maintained and calculated using the test results and the operating hours.

10.4.3 Opacity for the kiln shall not exceed 20%. **Opacity from the kiln is monitored with a continuous opacity monitor. At the time of inspection (11:50 am, 6/27), opacity was reading 4.9%.**

10.4.4 Emissions at the clinker cooler shall not exhibit 10% opacity, or greater. **Opacity from the clinker cooler is monitored with a continuous opacity monitor. At the time of inspection (11:50 am, 6/27), opacity was reading 1.99%.**

These PM and opacity standards apply at all times except during periods of startup, shutdown, and malfunction (40 CFR Part 60, Subpart A, 60.11(c)), however, at all times, the permittee shall use good operating practices as required by Condition 19 of this permit.

Continuous Opacity Monitor

The permittee shall install, calibrate, maintain, and operate in accordance with 40 CFR Part 60, Subpart A, 60.13 and Subpart F, a continuous opacity monitoring system to measure the opacity of emissions discharged into the atmosphere from the kiln and from the clinker cooler. (40 CFR Part 60, Subpart F)

A continuous opacity monitor is installed for the kiln and clinker cooler. Results at the time of inspection are listed above in 10.4.

The permittee shall submit reports of excess emissions. Periods of excess emissions that shall be reported are defined as all 6-minute periods during which the opacity limits listed above is exceeded. The content of these reports shall comply with the requirements of 40 CFR Part 60, Subpart A, 60.7(c). Such reports shall be submitted semiannually (40 CFR Part 60, Subpart F). Compliance with the reporting requirements of this section shall not relieve the owner or

operator of the reporting requirements found in General Condition 4 of Section IV of this Operating Permit.

Excess emission reports are sent to the Division for review.

Method 9 observations shall be used when the opacity monitoring system is unable to provide quality assured data. The observations shall be taken as follows: A Method 9 observation shall be performed by a certified opacity observer once per day. In addition, during these periods, the permittee shall maintain records of all instances when visible emissions are observed. These records shall include the date and time of the visible emissions, any circumstances pertinent to the emissions and any action taken as a result of the visible emissions. If visible emissions are present, a Method 9 observation shall be performed by a certified opacity observer at least once per hour while visible emissions are present. If any Method 9 opacity observation required above exceeds the applicable standard, additional observations must be performed. Consecutive observations shall be performed until two (2) consecutive observations are in compliance with the standard.

The plant has one person that is certified to take Method 9 opacity readings. To date, this has not been necessary as fugitive dust problems have been solved prior to needing to take readings.

The visual emissions records, records of results of Method 9 readings and a copy of the Method 9 reader's certification shall be kept on site and made available for inspection upon request. Copies of any observations exceeding the applicable standard shall be submitted with the next scheduled report.

Method 9 readings are kept for review and a copy of the reader's certification is kept in a binder for review.

The permittee shall maintain records of the occurrence and duration of any startup, shutdown, or malfunction in the operation of the kiln and clinker cooler; any malfunction of the air pollution control equipment; and any periods during which a continuous monitoring system or monitoring device is inoperative. (40 CFR Part 60, Subpart A, 60.7(b))

Records are maintained on startup, shutdown, or malfunction in the operation of the kiln and cooler. This information is kept in a logbook that is maintained/kept in the control room.

The permittee shall keep records of daily opacity measurements (Construction Permit 12BO444-2).

Daily opacity measurements are printed out and kept for review.

Note: On and after the MACT compliance date of June 10, 2002, any affected source subject to the provisions of the MACT standard is exempted from any otherwise applicable new source performance standard contained in 40 CFR part 60, Subpart F. (63.1356)

- 10.5 Particulate matter emissions from the kiln shall not exceed 0.275 pound per ton of feed (dry basis). (Incorporated directly into this operating permit through a combined construction/operating permit process, to make the annual PM emission limits enforceable.)

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 10.4, above.

Stack testing has been done and was done in accordance with condition 10.4.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of emissions calculations are maintained and calculated using the test results and the operating hours.

This PM standard applies at all times except during periods of startup, shutdown, and malfunction, however, at all times, the permittee shall use good operating practices as required by Condition 19 of this permit.

The permittee shall maintain records of the occurrence and duration of any startup, shutdown, or malfunction in the operation of the kiln and clinker cooler; any malfunction of the air pollution control equipment.

Records are maintained of malfunctions of the kiln or air pollution control equipment and kept in a logbook in the control room.

(Note: This Condition 10.5 remains in effect after the MACT compliance date.)

- 10.6 Opacity shall not exceed 20%, except as provided for under Condition 10.7. (Colorado Regulation No. 1, II.A.1).

Compliance with this opacity limit shall be demonstrated using the continuous opacity monitor as set forth in Condition 10.4.

Opacity has not exceeded 20%. Compliance with this limit is demonstrated with the continuous opacity monitor. At the time of inspection, opacity was 4.9% for the kiln and 1.99% for the cooler.

- 10.7 Opacity of emissions during fire building, cleaning of fire boxes, soot blowing, start-up, process modifications or adjustment of control equipment shall not exceed 30% for a period or periods aggregating more than six (6) consecutive minutes in any sixty (60) consecutive minutes (Colorado Regulation No. 1, Section II.A.4).

Compliance with this opacity limit shall be demonstrated using the continuous opacity monitor as set forth in Condition 10.4.

The readings from the continuous opacity monitor can be looked at and correlated to start-up or any work that was done on the equipment. This ensures opacity does not exceed 30% for more than six minutes.

- 10.8 Emissions of PM, PM₁₀, NO_x, SO₂, CO, and VOC shall not exceed the limits listed in the above table. (Construction Permit 12BO444-2, revised according to Section I, Condition 1.3, to delete the rolling twelve month total requirement).

Emissions have not exceeded the limits listed in the above table. See the table above for actual emissions.

NO_x, SO₂, CO, and VOC emissions shall be monitored on a continuous basis by certified continuous emission monitors. Proper installation, certification and operation shall include annual certification and quarterly audits of the calibration gases. (Construction Permit 12BO444-2). For APEN reporting and fee purposes, the CEM data shall be used to estimate emissions. When the CEMS are unable to provide quality assured data the following applies: When data is not available for 48 hours or less, the highest recorded hourly emission rate from the previous 24 hours shall be used. When data is not available for more than 48 hours, the maximum daily emission rate, based on the previous 30 days, shall be used.

NO_x, SO₂, CO, and VOC emissions are monitored with the CEM, which give a continuous reading to the control room. The reading is updated hourly and a new hourly average is calculated. The unit is calibrated quarterly. This data can be printed out and is stored so that at any time, records can be searched. A logbook of malfunctions is maintained and the unit calibrated on a quarterly basis. At the time of inspection, the readings from the CEM were: CO – 53.7 lbs/hour, NO_x – 558.7 lbs/hour, SO₂ – -1 lbs/hour, THC/VOC – 0.6 ppm/0.0 lbs/hr. The average lbs/hour emissions for the day of 6/27 were: 41 lbs/hour CO, 494 lbs/hour NO_x, and 0 lbs/hour for SO₂ and VOCs.

Compliance with the PM and PM₁₀ emission limits shall be monitored using the stack test results and baghouse maintenance procedures under Condition 10.3, and the records of production rates required under Condition 10.1. For APEN reporting and fee purposes, the stack test results and the actual production rates as recorded according to Condition 10.1 shall be used to estimate emissions.

PM and PM₁₀ emissions are monitored with the stack test. The stack test was completed in April. A copy of the results are attached.

- 10.9 Sulfur dioxide emissions shall not exceed 7 pounds per ton of material (including fuel) processed (Construction Permit 12BO444-2, Colorado Regulation No. 1, Section VI.3.f.). Note: This is a facility-wide emission limit - Compliance shall be monitored using the CEM data obtained under Condition 10.8, and actual material throughputs recorded under Conditions 10.1 and 10.3, the relevant information recorded for the dryer, and any other information necessary from any other sources emitting sulfur dioxide at this facility.

Sulfur dioxide emissions shall not exceed 7 pounds per ton of material for the entire facility. See table above for the actual emissions. This number is based on coal throughput and the total SO₂ emissions.

- 10.10 For APEN reporting and fee purposes, HCl emissions shall be estimated using actual clinker production and the emission factor listed in the table above (Portland Cement Association Report, September, 1996). Records of emission calculations shall be maintained for inspection upon request.

HCl emissions are estimated using actual clinker production and the emission factor. At the time of inspection this had not been calculated, but will be part of the database system Southdown is getting this summer. Emissions calculated in the table above by the clinker production and emission factor.

- 10.11 Emissions of lead shall not exceed the limits shown in the table above. Records of the annual emissions shall be kept and made available for inspection upon request (Construction Permit 12BO444-2).

Stack Testing

Performance testing for lead shall be performed once during the permit term, and at least once every five years thereafter, in accordance with the requirements and procedures set forth in EPA Test Method 12 forth in 40 CFR Part 60, Appendix A. Testing shall be performed for each proposed fuel type, except natural gas. No testing is required if natural gas is the only fuel used during the permit term. Alternatively, the permittee may test using the worst case lead emitting fuel, and shall then use this emission rate to estimate PM emissions from all fuels. A stack testing protocol shall be submitted for Division approval at least thirty (30) calendar days prior to performance of the test. No stack test shall be performed without prior written approval by the Division. The Division reserves the right to witness the test. The required number of copies of the compliance test results shall be submitted to the Division within forty-five (45) calendar days of the completion of the tests.

For monitoring compliance with the emission limits, and for APEN reporting and fee purposes, the stack test results and the actual production rates as recorded according to Condition 10.1 shall be used to estimate emissions.

Lead testing has been done for this permit term. The results are listed above in the table.

- 10.12 Emissions of lead shall not be such that emissions result in an ambient lead concentration exceeding 1.5 micrograms per standard cubic meter averaged over a one-month period (Colorado Regulation No. 8, Part C, Section I.B). The permittee submitted modeling results which indicate compliance with this applicable requirement. A copy of the modeling analysis and results shall be maintained and made available for inspection upon request.

Emissions of lead do not exceed 1.5 micrograms per standard cubic meter averaged over a one-month period. The stack test results of the lead testing are listed above and the permittee submitted modeling results which indicate compliance with this requirement.

- 10.13 This source shall utilize Reasonably Available Control Technology (RACT) for VOC emissions (Colorado Regulation No. 7, II.C). Operation of this kiln and clinker cooler as designed represents RACT. Any modification of the design shall require a new RACT determination and modification or reopening of this permit.

Operation of this kiln and clinker cooler as designed represents RACT. The permittee understands that any modification of the design shall require a new RACT determination and modification or reopening of this permit.

- 10.14 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT standard, as set forth in Conditions 24 of this permit.

This source will be subject to the MACT June 10, 2002.

11. **P009 - Clinker/Gypsum Silos and Weigh Feeders
S026 - S035 and S021 - S051 - Weight Feed Conveyors
S024 - Transfer
S038 - Separator
S039 - S041 - Silos and Weigh Feeders
Surge Bin
P010 - Sheltered Clinker Storage Stockpile with Underground Reclamation
S017 - Clinker Drop
S023 - Transfer**

S024A - A-Frame Building

P010A - Outdoor Clinker Storage, and Handling of Clinker with Front End Loaders

S024B

P011 - Cement Finish Mill and Bucket Elevator/Auxiliaries

S036 & S037 - Grinding

P012 - High Efficiency Separator (Classifier) and Baghouse Collector

P013 - Cement Storage Silos/Packhouse/Loadout

S043 to S048 - Cement Silos

S049 & S050 - Packhouse

Two Bulk Cement Loadout Spouts

Parameter	Limitations	Emission Factors	Actual Process/Emissions
Process Limits	<i>P009</i> Clinker Handled: 600,000 tons/year 75,000 tons/month 4,000 tons/day		<i>P009</i> Clinker Handled: 435,795 tons/year 36,316 tons/month 1,585 tons/day
	----- <i>P010</i> Clinker Handled: 600,000 tons/year 75,000 tons/month 5,500 tons/day		----- <i>P010</i> Clinker Handled: 271,996 tons/year 22,666 tons/month 989 tons/day
	----- <i>P010A</i> Maximum Clinker Stored: 120,000 tons Clinker Handled: 180,000 tons/year 5,500 tons/day		----- <i>P010A</i> Maximum Clinker Stored: 45,175 tons Clinker Handled: 142,925 tons/year 519.73 tons/day
	----- <i>P011</i> Overall Fresh Feed to Mill: 600,000 tons/year 75,000 tons/month 3,000 tons/day		----- <i>P011</i> Overall Fresh Feed to Mill: 435,795 tons/year 36,316 tons/month 1,585 tons/day
	----- <i>P012</i> Cement Produced: 600,000 tons/year 75,000 tons/month 3,000 tons/day		----- <i>P012</i> Cement Produced: 435,795 tons/year 36,316 tons/month 1,585 tons/day
	----- <i>P013</i> Cement Handled: 600,000 tons/year 75,000 tons/month 4,000 tons/day		----- <i>P013</i> Cement Handled: 485,234 tons/year 40,436 tons/month 1,666 tons/day

Operating Hours	8,064 hours/year		5,287 hours/year
PM	<i>P009</i> 7.02 tons/year	0.03 gr/dscf	<i>P009</i> 2.64 tons/year
	<i>P010</i> 0.82 ton/year	0.03 gr/dscf	<i>P010</i> .332 tons/year
	<i>P010A</i> 3.35 tons/year		<i>P010A</i> 2.64 tons/year
	<i>P011</i> 15.93 tons/year	0.03 gr/dscf	<i>P011</i> 3.09 tons/year
	<i>P012</i> 35.86 tons/year	0.01 gr/dscf	<i>P012</i> 2.8 tons/year
	<i>P013</i> 11.39 tons/year	0.03 gr/dscf	<i>P013</i> 0.25 tons/year
PM₁₀	<i>P009</i> 3.51 tons/year 0.33 tons/month 22 lbs/day	50% of PM	<i>P009</i> 1.67 tons/year 0.13 tons/month 13.6 lbs/day
	<i>P010</i> 0.41 ton/year 0.05 ton/month 7.5 lbs/day	50% of PM	<i>P010</i> 0.22 tons/year 0.02 tons/month 1.9 lbs/day
	<i>P010A</i> 1.30 tons/year 0.16 ton/month 78 lbs/day		<i>P010A</i> 1.04 tons/year 0.08 tons/month 5.71 lbs/day
	<i>P011</i> 7.96 tons/year 0.71 ton/month 48 lbs/day	50% of PM	<i>P011</i> 1.82 tons/year .15 tons/month 16.44 lbs/day
	<i>P012</i> 17.93 tons/year 1.61 tons/month 107 lbs/day	50% of PM	<i>P012</i> 1.19 tons/year .13 tons/month 10.82 lbs/day
	<i>P013</i> 5.70 tons/year 0.72 ton/month 43 lbs/day	50% of PM	<i>P013</i> 0.63 tons/year 0.07 tons/month 6.4 lbs/day

- 11.1 The amount of clinker handled shall not exceed the limits listed in the table above. Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. (Construction Permit 98BO0259, revised according to Section I, Condition 1.3, to delete the rolling twelve month total requirement). Records of the monthly and annual handling rates shall be kept on-site and made available for Division review upon request. The permittee shall maintain records of the number of days of operation for each month. Compliance with the daily limits shall be monitored by dividing the monthly handling rates by the monthly number of days of operation. Records of the number of days of operation and the daily calculation rates shall be maintained and made available upon request.

Amount of clinker shall not exceed the limits above. Compliance is demonstrated on a monthly basis. Records are maintained and available for review. See table above for actual throughputs which include the months of April 1999 through December 1999. Records were not maintained until that point.

- 11.2 Annual hours of operation shall not exceed 8,064. (Construction Permit 98BO0259, as revised according to Section I, Condition 1.3, to delete the rolling twelve month total requirement). Records of the annual hours of operation shall be kept on-site and made available for Division review upon request.

Annual hours of operation have not exceeded 8,064. Records of hours of operation are maintained. See table above for actual hours.

- 11.3 PM and PM₁₀ emission rates shall not exceed the limits listed in the table above. Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. (Construction Permit 98BO0259, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve month total requirement). For all sources except P010A, compliance with the emission limits shall be monitored using the results of the test required in this Condition 11.3, the maximum design flow rate of the control devices, the actual hours of operation as recorded under Condition 11.3, and the baghouse maintenance and operating procedures required in this Condition 11.3. Compliance with the daily PM₁₀ emission limits shall be determined using the same information, as well as the number of monthly operating days. The Annual Certification shall describe how the inspection, maintenance work, and repair documentation required in Condition 11.2 affected emissions. For P010A, compliance with the emission limits shall be demonstrated by meeting the production limits set forth in Condition 11.1, and by applying the Fugitive Particulate Emission Control Plan set forth in Condition 11.7.

PM and PM₁₀ emission rates shall not exceed the limits listed in the table above. Compliance is determined on a monthly basis and determined by meeting the production limits in Condition 11.1. Points 009, 010, 011, 012 and 013 emissions are based on throughputs for April 1999 through December 1999 - the date permit tracking was required. Point 010A is based on a full year. Emissions are well below permit limits and even if the numbers were doubled, emissions would still be below permit limits. Emissions for all points except P010A were calculated with stack test results from September 1999. P010A calculated by emission factors. For 2000, tracking of throughputs is done monthly and to date, they are under their permit limits. See table above for actual emissions.

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 23 of this permit, except that tests are required once every five years (once per permit term), instead of annually. A test shall be performed for a representative bagfilter for each activity group (P009, P010, P011, P012, and P013). At least one activity group will be tested within one year of permit issuance to include a PM₁₀ fraction analysis to verify the assumption that PM₁₀ emissions are 50% of total PM. (Construction Permit 98BO0259) If the stack test indicates that PM₁₀ emissions exceed 50% of total PM, the Division will need to review the activities permitted under Construction Permit 98BO0259 for non-attainment NSR applicability. The permittee may perform a PM₁₀ fractional analysis (test) more frequently if this is needed to obtain representative information. If the permittee elects to perform additional analyses due to the failure of the initial test to verify that PM₁₀ emissions are less than 50%, approval for, and the schedule for, performing such

additional tests shall be obtained from the Division. If a stack test has been performed to meet the Construction Permit stack test requirement, no further testing is required during this initial permit term.

Stack testing was done for points 009, 010, 011, 012 and 013 in September 1999. for this permit term. A test of bagfilters has also been done for particulate matter. Five of the finish mill baghouses have been tested and have show they are in compliance. See attached copy of test results. See table above for actual emissions.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouse shall be conducted in accordance with manufacturer's specifications and the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request. (Construction Permit 98BO0259)

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of emissions calculations are maintained and calculated using the test results and the operating hours.

For all sources except P010A, for APEN reporting and fee purposes, annual PM emissions shall be calculated using the using the results of the test required in this Condition 11.3, the maximum design flow rate of the control devices, and the actual hours of operation as recorded under Condition 11.2. PM 10 emissions shall be assumed to be 50% of PM emissions, unless the stack test required under this Condition 11.3 shows otherwise. For P010A, actual throughput rates recorded under Condition 11.1, appropriate emission factors and equations, and control efficiencies shall be used to estimate emissions. Records of the emissions calculations shall be maintained and made available for inspection upon request.

See above table and attached emissions for P010A.

- 11.4 These sources are subject to the requirements of 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants as set forth in Condition 21 of this permit. Note: On and after the MACT compliance date of June 10, 2002, any affected source subject to the provisions of the MACT standard is exempted from any otherwise applicable new source performance standard contained in 40 CFR part 60, Subpart F. (63.1356)

This source is subject to the requirements of 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants as set forth in Condition 21 of this permit. After June 10, 2002 they will be subject to the MACT standard for Portland Cement.

- 11.5 These sources, except for P010A, are subject to the opacity limits set forth in Condition 22 of this permit.

These sources are subject to the opacity limits set forth in Condition 22. At the time of inspection, no opacity over 20% was noted. Compliance with this condition is monitored with the routine bagfilter operation and maintenance which was described above in Condition 11.3.

- 11.6 The following particulate emission control measures shall be applied to particulate emission producing sources (Construction Permit 98BO0259).

- 11.6.1 Roads and edges of clinker pile(s) shall be watered, except during below freezing temperatures, at least once per day, or more often, as necessary to abate fugitive particulate emissions. If, at any time, the opacity of particulate emissions from the clinker pile(s) exceeds ten (10) percent (as required under NSPS, Subpart F), then the pile(s) shall be watered at least once per day until a crust forms on the surface. **Roads and clinker piles is watered at least once a day as part of the regular plant watering and more often if needed. Records are maintained of the gallons of water used.**
- 11.6.2 Traffic on and around storage pile(s) shall be minimized. **Traffic on and around the storage piles is minimized.**
- 11.6.3 Height of fall material shall be minimized during front-end loader handling. (Construction Permit 98BO0259, as modified according to Section I, Condition 1.3 of this permit) **Height of fall for the front-end loader is minimized.**
- 11.6.4 Vehicle traffic on unpaved surfaces shall be restricted to established roadways. **Vehicle traffic is restricted to paved areas.**
- 11.6.5 Clinker shall be reclaimed from the storage pile(s) as soon as possible, consistent with seasonal product demands and covered storage constraints. (Construction Permit 98BO0259, as modified according to Section I, Condition 1.3 of this permit) **Clinker is reclaimed from storage as soon as possible.**
- 11.6.6 Paved areas shall be swept regularly with a high efficiency vacuum sweeper to minimize material buildups. In addition, these areas will be watered as necessary and vehicle traffic suspended or rerouted if fugitive emissions become a concern. (Construction Permit 98BO0259, as modified according to Section 1, Condition 1.3 of this permit) **Paved areas are swept daily. The plant has two high efficiency vacuum sweepers. A smaller one gets underneath the kiln, etc and the larger one does the paved road areas.**
- 11.6.7 Activities causing fugitive dust emissions shall be suspended when wind speeds reach or exceed 30 miles per hour, averaged over a 60-minute period. Only those activities affected by wind speed, and for which it is possible to "suspend operation" need be shut down (i.e., the permittee cannot "shut down" storage piles, thus this condition would not apply to storage piles). Activities may continue when the average wind speed drops below 30 m.p.h. (Construction Permit 98BO0259, as modified per Section 1, Condition 1.3 of this permit) **Activities are suspended when the winds reach or exceed 30 mph averaged over a 60-minute period. The wind speed instrument is located at the Dowe Flats site and sends updated information to the control room.**
- 11.6.8 The drop of clinker to the outside chute shall be managed with a seal between the hopper and material during operation to allow effective operation of the dust collector such that emissions are minimized. (Construction Permit 98BO0259, as modified according to Section 1, Condition 1.3 of this permit) **The drop of clinker to the outside chute is managed with a seal between the hopper and material during operation to minimize emissions. Seal is in place and emissions are minimized.**

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when average wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h., averaged over a sixty minute period. The Annual Certification shall include a statement that the activities identified in Condition 11.6.7 were suspended when average wind speeds reached or exceeded the 30 m.p.h. threshold.

The wind speed instrument is located at the Dowe Flats site and sends updated information to the control room. This information can be printed for any time period and the annual

certification will include a statement that activities were suspended when the wind speed reached the threshold.

A weekly inspection of the site shall be conducted to ensure the emission control elements are in place and effective. In addition, at any time when a fugitive dust problem is observed, the permittee shall take action to correct the problem. The permittee shall maintain records of the date and time of any fugitive dust problem observed, and the type and time of action taken to correct the problem. These records shall be maintained on site for inspection upon request. The Annual Certification shall include a statement that the PM Emission Control Plan measures are in place and effectively minimizing fugitive emissions.

A daily walk through of the plant is conducted to check for fugitive emissions. Fugitive dust problems are noted on the observation sheet with the time and date and a photo.

Southdown needs to document better when the problem was fixed and what the correction was. Steve stated they will begin doing this. The inspection sheets are maintained on-site in a binder and available for review.

- 11.7 The High Efficiency Separator (Classifier) and Baghouse Collection System (P031) shall be maintained under a negative pressure (Construction Permit 98BO0259).
The gauge to measure the pressure can be seen in the console room. Negative pressure is needed on these units in order for them to operate.

- 11.8 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT standard, as set forth in Conditions 24 of this permit.
This source will be subject to the MACT on June 10, 2002.

12. **P014 - Clinker and Fuel Handling**
S020 - Coal Silo
S025 - Coal/Clinker Hopper

Parameter	Limitations	Emission Factors	Actual Process/Emissions
Process Rate			101,869 tons/coal received
PM & PM ₁₀		PM & PM ₁₀ : 2.9 x 10 ⁻⁵ lb/ton (for each baghouse stack)	29.5 lbs/year (for each baghouse stack)
PM	See Condition 12.3		See 12.3

- 12.1 Records of the annual amounts of material throughput shall be maintained and made available to the Division for inspection upon request.
Records are maintained and available for inspection on request. Numbers are based on coal received for 1999. See table above for actual material throughput.
- 12.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual material throughputs recorded under Condition 12.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). This factor accounts for control equipment

efficiency. Records of the calculations shall be maintained and made available for inspection upon request.

Annual emissions are estimated using the throughputs and emission factor listed above. See table above for actual emissions.

- 12.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed:

$$E = 17.31 (P)0.16$$

Where:

E is the allowable particulate emissions in lbs/hr.

P is the process weight rate in tons/hr

(Colorado Regulation No. 1, III.C.1.b). In absence of evidence to the contrary, compliance with this limit shall be monitored using the baghouse operating and maintenance requirements of Condition 22.

Compliance with this limit is monitored using the baghouse operating and maintenance requirements as per Condition 22 of this permit. Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request.

- 12.4 These sources are subject to the opacity limits set forth in Condition 22. of this permit. **These sources are subject to the opacity limits set forth in Condition 22. At the time of inspection, no opacity over 20% was noted. Compliance with this condition is monitored with the routine bagfilter operation and maintenance which was described above.**
- 12.5 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT standard, as set forth in Conditions 24 of this permit.
This source will be subject to the MACT on June 10, 2002.

13. **P007A - Handling and Processing of CKD And Raw Material Waste Dust**
040A - Conveyance of CKD to Silo A5 or 2844
040B - Conveyance of Benefication Dust to Silo 2844
041 - Pug Mill
042 - Truck Hauling to Lyons Quarry

Parameter	Limitations	Emission Factors	Actual Process/Emissions
Process Rate	040A & B, 041 Total Quantity of Materials Annually conveyed, CKD and benefication dust, together, on a dry basis: 133,000 tons/year 16,625 tons/month		040A & B 041 CKD and benefication dust on a dry basis: 30,755.93 tons/year 2,562.99 tons/month 145.22 tons/day (average - all tons/day calculations have been

	600 tons/day <i>042</i> Total Quantity of material Hauled and Disposed, on Wet Basis: 173,403 tons/year 21,675 tons/month 800 tons/day		under permit limit) <i>042</i> Wet Basis: 40,099 tons/year 3,341.5 tons/month 189.34 tons/day (avg.)
PM	<i>040A&B</i> 5.30 tons/year <i>041</i> 2.66 tons/year <i>042</i> 5.50 tons/year	0.03 gr/dscf 0.8 lb/ton 	<i>040A&B</i> .048 tons/year <i>041</i> .615 tons/year <i>042</i> Not Calculated, but since under production limits, assume emissions are under as well.
PM₁₀	<i>040A & B</i> 2.65 tons/year 0.33 tons/month 24 lbs/day <i>041</i> 2.66 tons/year 0.33 tons/month 24.00 lbs/day <i>042</i> 2.50 tons/year 0.31 tons/month 23 lbs/day	50% of PM 0.2 lb/ton 	<i>040A&B</i> .024 tons/year .002 tons/month .24 lbs/day <i>041</i> .615 tons/year .05 tons/month 5.8 lbs/day (avg.) <i>042</i> Not Calculated, but since under production limits, assume emissions are under as well.

- 13.1 Process rates shall not exceed the maximum rates specified in the table above. Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. Records of the monthly and annual process rates shall be maintained and made available for inspection upon request. (Construction Permit 98BO0315, revised according to Section I, Condition 1.3 of this permit, to delete the rolling twelve month total requirement) The permittee shall maintain records of the number of operating days for each month. Compliance with the daily process rates shall be monitored by dividing the monthly process rate by the number of days of operation. Records of the number of days of operation and the process rate calculations shall be maintained and made available for inspection upon request.
- Process rates have not exceeded the limits listed in the table above. See table above for actual process rates. Records are maintained on a monthly basis with the days of operation and available for inspection.**

- 13.2 PM and PM₁₀ emission rates shall not exceed the limits listed in the table above. Compliance with the monthly limits shall be demonstrated until the beginning of the first calendar year after commencement of operation at the new production rate. Thereafter, compliance with the annual limits shall be determined on an annual basis. (Construction Permit 98BO0315, revised according to Section I, Condition 1.3, to delete the rolling twelve month total requirement) For 040A&B, compliance with the emission limits shall be monitored using the results of the test required in this Condition 13.2, the maximum design flow rate of the control devices, the monthly hours of operation, and the baghouse maintenance and operating procedures required in this Condition 13.2. Compliance with the daily PM₁₀ emission limits shall be determined using this information and the number of days of operation. The Annual Certification shall describe how the inspection, maintenance work, and repair documentation required in Condition 13.2 affected emissions. For 041, compliance with the emission limits shall be monitored by meeting the production limits set forth in Condition 13.1 and the baghouse maintenance and operating procedures. For 042, compliance with the emission limits shall be monitored by meeting the production limits and by applying the Fugitive Particulate Emission Control Plan set forth in Condition 13.6.

PM and PM₁₀ emission rates have not exceeded the rates in the table above. See table above for the actual emissions. Emissions calculated for points 0410A and B from stack test results. Emissions for 041 calculated with emission factors. Point 042 compliance is determined by production rates, the hours of operation and by following the baghouse operation and maintenance procedures. Annual certification will verify the above.

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 23 of this permit, except that a test is required once every five years (once per permit term), instead of annually. For 40A&B, performance testing shall be performed at a representative baghouse. The first test (performed within one year of permit issuance) shall include a PM₁₀ fraction analysis to verify the assumption that PM₁₀ emissions are 50% of total PM. (Construction Permit 98BO0315) If stack test results indicate PM₁₀ emissions represent more than 50% of total PM, the Division will need to review the activities permitted under 98BO0315 for non-attainment NSR applicability. The permittee may perform a PM₁₀ fractional analysis (test) more frequently if this is needed to obtain representative information. If the permittee elects to perform additional analyses due to the failure of the initial test to verify that PM₁₀ emissions are less than 50%, approval for, and the schedule for, performing such additional tests shall be obtained from the Division.

Stack testing for these points has been completed for this permit term and was conducted in January, 2000. Performance testing has been done on the baghouses. See attached report.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request. (Construction Permit 98BO0315)

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of

emissions calculations are maintained and calculated using the test results and the operating hours.

For 40A&B, for APEN reporting and fee purposes, annual PM emissions shall be calculated using the test results, the maximum design flow rate of the control device, and hours of operation. PM10 emissions shall be assumed to be 50% of PM emissions, unless stack test results indicate otherwise. For 41, annual emissions shall be calculated using the actual process rates, the emission factor listed in table above (Construction Permit 98BO0315), and appropriate control equipment efficiencies. Note: The analysis for this operating permit used control efficiencies of 95% for PM and 80% for PM10 for 41. For 42, actual throughput rates recorded under Condition 13.1, appropriate emission factors and equations, and control efficiencies shall be used to estimate emissions. Records of the emissions calculations shall be maintained and made available for inspection upon request.

Emissions calculated by stack tests and emission factors. See above table for actual emissions.

- 13.3 These sources are subject to the requirements of 40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants as set forth in Condition 21 of this permit.
This source is subject to 40CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants. Opacity/stack test on CKD baghouse was completed in January 2000. Results were 0% opacity. See attached copy of results. See Condition 21 for specifics.
- 13.4 These sources, except for hauling and disposal, are subject to the opacity limits set forth in Condition 22 of this permit.
Sources are subject to the opacity limits in Condition 22. No opacity was noted at the time of inspection.
- 13.5 The following particulate emission control measures shall be applied to particulate emission producing sources (Construction Permit 98BO0315)
- 13.5.1 Transfer points shall be enclosed with the exception of the drop from the pugmill to the truck. Emissions shall be controlled in the drop to the truck through effective moisture control in the pugmill. (Construction Permit 98BO0315, as modified per Section 1, Condition 1.3 of this permit) **Transfer points (except the pugmill to truck) are enclosed. The pugmill ensures the material has enough moisture so that fugitive emissions are controlled.**
- 13.5.2 Moisture content of the materials prior to transfer to pug mill shall be adequate to effectively control emissions. **Moisture is adequate in materials prior to the pugmill. Material is watered if necessary.**
- 13.5.3 Haul roads shall be treated with chemical dust suppressants, as often as required, to maintain a surface crust, and adequately watered to achieve a minimum control efficiency of 80%. Records of such application of dust suppressants and watering shall be maintained at the site. **Haul roads are treated with chemical dust suppressants as part of the entire plant effort to control fugitive dust. Records of these applications are on file for review. In addition, a water truck treats the haul roads at least once a day.**
- 13.5.4 At the disposal pit, the material shall be compacted and stabilized to minimize emissions. **Material is compacted to minimize emissions.**
- 13.5.5 CKD exposed at the disposal site, shall be minimized, and no more than seven (7) acres shall be exposed or used for disposal at a time. Fugitive particulate matter emissions

from areas not being actively used shall be minimized using a shale cover.(Construction Permit 98BO0315, as modified per Section 1, Condition 1.3 of this permit) **Exposed CKD is minimized with no more than 7 acres exposed or used for disposal. Once an area is filled, it is capped with a shale cover to minimize emissions.**

- 13.5.6 A sprinkler system shall be in operation as necessary to effectively control emissions at the disposal area. Emissions shall be controlled during dumping by effective moisture control during pugging. (Construction Permit 98BO0315, as modified per Section I, Condition 1.3 of this permit) **A sprinkler system is in operation at the disposal pit area. The pugging process makes the material moist for dumping into the disposal pit. The sprinkler system is on a timer so every ten minutes some section of the sprinkler is operating to keep the material moist. At the time of inspection a section of the sprinkler system was operating and no fugitive dust was noted from the pit.**
- 13.5.7 Haul trucks of 95 tons capacity or greater shall be used to minimize the vehicle-miles traveled. Spillage and exposure to wind shall be minimized by restricting the material load to 75 percent of the volume capacity of the trucks. (Construction Permit 98BO0315, as modified per Section I, Condition 1.3 of this permit.) **Only large 95 ton capacity trucks are used so that miles traveled are minimized and trucks are never filled completely to reduce emissions. Actually, Southdown does not have any haul trucks that have a lower capacity.**
- 13.5.8 Spillage and accumulations of particulate matter on roadways shall be watered, cleaned up regularly, and managed to insure they do not contribute to fugitive emissions during operation. Vehicle traffic will be suspended or rerouted from spillage areas as necessary to minimize fugitive emissions.(Construction Permit 98BO0315, as modified per Section 1, Condition 1.3 of this permit) **Spillage on roads are cleaned with the sweeper trucks and by hand if needed. Roadways are watered as part on plant fugitive dust control plan.**
- 13.5.9 Compliance inspection of the CKD pugging, hauling, and storage operations shall be conducted at least 2 times per shift (at no less than 2 hour intervals). Records of such inspections shall be maintained at the site.(Construction Permit 98BO0315, as modified per Section 1, Condition 1.3 of this permit) **This is checked as part of the daily opacity walkthrough of the plant. Item #12 - Waste Dust Loadout indicates the two times during the day this is inspected. See attached copy of Daily Opacity Observation Sheet.**
- 13.5.10 Activities causing fugitive dust emissions shall be suspended when wind speeds reach or exceed 30 miles per hour, averaged over a 60-minute period. Only those activities affected by wind speed, and for which it is possible to "suspend operation" need be shut down (i.e., the permittee cannot "shut down" storage piles, thus this condition would not apply to storage piles). Activities may continue when the average wind speed drops below 30 m.p.h. (Incorporated directly into this operating permit per Section 1, Condition 1.3 of this permit) **Activities that cause fugitive dust emissions are suspended when wind speeds reach or exceed 30 miles per hour, averaged over a 60 minute period.**

Records of dust suppressant use shall be maintained and made available upon request. The Annual Certification shall indicate that the PM Emission Control Plan measures are in place and effectively minimizing fugitive emissions.

Records of applications of dust suppressants are maintained.

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when average wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h., averaged over a sixty minute period. The Annual Certification shall include a statement that the activities identified in Condition 11.6.7 were suspended when average wind speeds reached or exceeded the 30 m.p.h. threshold.

The anemometer is installed at the Dowe Flats site and sends information directly back to the control room. Information is stored on the computer and any time frame can be pulled up to see what the wind speed was.

- 13.6 Only Pit "C" shall be used for disposal of CKD and Benification Dust. The one-way haul distance is 0.38mile. (Construction Permit 98BO0315).

Only Pit "C" is used for CKD and dust disposal.

14. P016 - Haul Roads

- 14.1 Records of the annual amount of material hauled, and all other information necessary to estimate emissions, shall be maintained and made available to the Division for inspection upon request.

A description of the activity and the units of measure for each activity is used to estimate emissions for this point. See attached summary.

- 14.2 For APEN reporting purposes, annual PM and PM10 emissions shall be estimated using the records obtained under Condition 14.1, and appropriate emissions factors and/or equations and control efficiencies. Records of the calculations shall be kept on site for Division inspection upon request.

Emissions are calculated as described above. See attached.

NOTE: Measures to minimize fugitive emissions from the haul roads are included in the Particulate Matter Emission Control Plans listed in other Conditions of this Operating Permit.

15. T001 - Gasoline Storage Tank 3,000 Gallon Capacity

- 15.1 The permittee shall maintain records of the annual amount of gasoline throughput. Records shall be made available for inspection upon request.

Records of the gasoline throughput are maintained. For 1999, 14,005 gallons was used.

- 15.2 For APEN reporting and fee purposes, annual VOC emissions shall be estimated using the actual throughput rate and the EPA's TANKS program, version 2.0 or higher. Records of the emission calculations shall be made available for inspection upon request.

This point was previously grandfathered so emissions have not been calculated. Southdown is working on getting a spreadsheet/database program that will calculate this information for them. At the time of inspection, this database was near completion.

- 15.3 This source is subject to the Reasonably Available Control Technology (RACT) requirements set forth in Colorado Regulation No. 7, VI.B.3, as follows.

15.3.1 The permittee shall not allow the transfer of petroleum liquid from any delivery vessel into this tank unless the tank is equipped with a submerged fill pipe and the vapors displaced from the storage tank during filling are processed by a vapor control system. A

vapor balance system shall be deemed "approved" if its design and operation are in accordance with the applicable requirements of Appendices A and B of Colorado Regulation No. 7. Tanks equipped with a submerged fill pipe shall meet the specifications of Appendix A of Colorado Regulation No. 7. **Submerged fill was installed in 1996 on the petroleum transfer units.**

- 15.3.2 The vapor control system shall include one or more of the following:
- (i.) A vapor-tight line from the storage tank to delivery vessel (i.e. an approved control system).
 - (ii) A refrigerator-condensation system or equivalent designed to recover at least 90 percent by weight of the organic compounds in the displaced vapor.

System is equipped with a vapor tight line.

- 15.3.3 The permittee shall ensure that operating procedures are used so that gasoline cannot be transferred into the tank unless the vapor control system is in use. **If the line is not connected correctly, gas will not be transferred.**

- 15.3.4 The vapor balance system shall meet the specifications of Appendix A of Colorado Regulation No. 7. **System meets the specification of Regulation No. 7.**

- 15.3.5 The vapor balance system and the vapor control system shall meet the requirements of Section XV of Colorado Regulation No. 7. **System meets the specification of Regulation No. 7.**

16. Opacity - Daily Plant Walk Through

A daily plant walk through shall be performed to look for visible emissions. During the walk through, an observer will survey the plant from 5 observation points. From these 5 locations, all of the facility's baghouses and material transfer points can be observed. The results of these observations shall be logged on a daily observations sheet. If visible emissions in excess of typical emissions are observed from any stack, the permittee shall undertake the appropriate corrective process and/or maintenance actions. When these actions are completed, that stack will be observed again. If visible emissions in excess of typical emissions remain, the permittee shall perform a Method 9 test of that stack. If any of the method 9 opacity observations required above exceed the applicable standard, additional observations must be performed. Consecutive observations shall be performed until two (2) consecutive observations are in compliance with the standard.

A daily plant walk-through is conducted to check for visible emissions. This was started in February 2000. The plant is surveyed from five points where one can see all the facilities' baghouses and transfer stations. A picture of the plant is taken and attached to the daily observation sheet. Opacity problems are corrected and to date Southdown has not had to perform a Method 9 reading.

The corrective actions taken and the follow-up observations shall be noted on the daily observation sheet. **Not all the records are clear about what actions were taken and when the problem was corrected. I mentioned this to Steve and he said they would be better about documenting this.**

If visible emissions from fugitive sources are noted, the permittee shall investigate to insure that the provisions of the appropriate fugitive dust control plan are being implemented, and shall note on the observation sheet, along with additional actions taken.

Problems are noted and corrected, again the sheet is not always very clear about what was done, but this will be corrected.

The daily observation sheets shall be maintained and made available for inspection upon request.
Daily observation sheets are maintained and kept in a binder for inspection upon request.

These procedures do not apply to the kiln and cooler, which are equipped with opacity monitors.
The daily walkthrough is for the entire plant except for the kiln and cooler since they have continuous opacity monitors.

17. Insignificant Activities

The permittee shall at least annually inspect and determine whether the categorically insignificant activities are in compliance with the opacity standard and are exempt from the current regulatory APEN reporting requirements. The permittee shall record in a log the results of this inspection.

Insignificant activities will be reviewed annually for compliance with opacity and if they are still exempt from an APEN.

18. Reporting

All reports shall be postmarked within the first thirty (30) days immediately following the end of the reporting period, unless a different response time is identified elsewhere in this permit. The compliance monitoring report shall be in the format identified in Appendix B of this permit.

The compliance monitoring report will follow the format in Appendix B. The semi-annual was due August 1st, 2000 and the annual is due March 2001. Per Louis McBride at APCD, the semi-annual report has NOT been submitted. However, Southdown has a receipt from their overnight courier showing that the report was signed for on July 19th, 2000. Steve stated he sent it directly to Long Nguyen.

19. Good Operating Practices

At all times, including periods of startup, shutdown, and malfunction, owners and operators shall, to the extent practicable, maintain and operate any affected facility including associated air pollution control equipment in a manner consistent with good air pollution control practice for minimizing emissions. Determination of whether acceptable operating and maintenance procedures are being used will be based on information available to the Division which may include, but is not limited to, monitoring results, opacity observations, review of operating and maintenance procedures, and inspection of the source. (40 CFR Part 60, 60.11(d)).

Although this source has taken many positive steps to control fugitive emissions from the plant and have worked hard with the North Boulder Environmental Health Community Task Force (NBEHCTF), there is still room for improvement. At the time of inspection, roads were adequately watered, however, under parts of the kiln and cooler, piles of cement dust could be seen. This area is swept/shoveled by hand since the sweeper can not get under the equipment. These areas should be cleaned more frequently since it is source of fugitive dust when it gets windy.

No article, machine, equipment or process shall be used to conceal an emission which would otherwise constitute a violation of an applicable standard. Such concealment includes, but is not limited to, the use of gaseous diluents to achieve compliance with an opacity standard or with a standard which is based on the concentration of a pollutant in the gases discharged to the atmosphere. (40 CFR Part 60, Subpart A, 60.12)

Nothing is used to conceal emissions.

20. Opacity and PM Limits for Subpart OOO Sources

40 CFR Part 60, Subpart OOO, Standards of Performance for Non-Metallic Mineral Processing Plants, as adopted by reference in Colorado Regulation No. 6, Part A, limits emissions as follows:

- 20.1 The permittee shall not cause to be discharged into the atmosphere any stack emissions which exhibit greater than 7 percent opacity.

This condition applies to the transfer points and crushers at the plant. Each transfer point is equipped with a baghouse that is maintained per the manufacturer's instructions. A plant walkthrough is conducted daily for opacity and problems are noted and repaired. These records are kept in a binder and available for review. At the time of the follow-up inspection, no opacity was noted.

Monitoring

An EPA Method 9 opacity observation, in accordance with Appendix A of 40 CFR Part 60, and 60.675(c), shall be conducted for the bagfilter at least semi-annually. In addition, the procedures set forth under Condition 16 of this permit shall be followed.

A Method 9 opacity observation has been done for 2000 by APT.

Records of results of semiannual Method 9 readings and a copy of the Method 9 reader's certification shall be kept on site and made available to the Division upon request. Copies of any observations exceeding the applicable standard shall be submitted with the next scheduled report.

Records are maintained and kept on site. Copies of the readings and the reader's certification number are kept in a binder and available for review. Currently, one person at the plant is certified to do a Method 9.

- 20.2 The permittee shall not cause to be discharged into the atmosphere any stack emissions which contain particulate matter in excess of 0.05 gram per dry standard cubic meter.

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 23 of this permit, except that tests are required once every five years (once per permit term), instead of annually. In addition, the permittee need only test one representative baghouse stack for the conveyor.

Stack testing for this permit term has been completed. See attached report.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of emissions calculations are maintained and calculated using the test results and the operating hours.

This PM and opacity standard apply at all times except during periods of startup, shutdown, and malfunction (40 CFR Part 60, 60.11 (c)), however, at all times, the permittee shall use good operating practices as required by Condition 19.0 of this permit. The permittee shall maintain records of the occurrence and duration of any startup, shutdown, or malfunction in operation of this source; and any malfunction of the air pollution control equipment (40 CFR Part 60, Subpart A, 60.7(b), as adopted by reference in Colorado Regulation No. 6, Part A).

Records are maintained on the occurrence and duration of any startup, shutdown or malfunction.

21. Opacity Limits for Subpart F Sources

40 CFR Part 60, Subpart F, Standards of Performance for Portland Cement Plants, as adopted by reference in Colorado Regulation No. 6, Part A, limits emissions as follows.

Gases discharged into the atmosphere shall not exhibit 10 percent opacity, or greater.

This condition applies to the conveyor, underground reclamation system, front end loader activity, storage and handling of raw materials, the raw material dryer, and the kiln and cooler. The kiln and the cooler are equipped with an opacity monitor that gives a continuous reading to the control room. At the time of the follow-up inspection, the opacity was 4.9% and 1.99% respectively. No emissions were seen from the transfer points or baghouses at the plant.

This opacity standard applies at all times except during periods of startup, shutdown, and malfunction (40 CFR Part 60, 60.11 (c)), however, at all times, the permittee shall use good operating practices as required by Condition 19. of this permit. The permittee shall maintain records of the occurrence and duration of any startup, shutdown, or malfunction in operation of this source; and any malfunction of the air pollution control equipment (40 CFR Part 60, Subpart A, 60.7(b), as adopted by reference in Colorado Regulation No.6, Part A).

Records are maintained on the occurrence and duration of any startup, shutdown or malfunction.

Monitoring

Except for the kiln and cooler, compliance with this limit shall be monitored as follows: An EPA Method 9 opacity observation, in accordance with Subpart F requirements, shall be conducted at least semi-annually. In addition, the procedures set forth under Condition 16 of this permit shall be followed. Records of results of semiannual Method 9 readings and a copy of the Method 9 reader's certification shall be kept on site and made available to the Division upon request. Copies of any observations exceeding the applicable standard shall be submitted with the next scheduled report.

Compliance this limit is determined by a daily plant walkthrough for opacity. Records are maintained and kept on site for review. Method 9 readings were completed earlier this year for this permit condition. Records of the Method 9 readings are kept in a binder for review.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of

emissions calculations are maintained and calculated using the test results and the operating hours.

Note: On and after the MACT compliance date of June 10, 2002, any affected source subject to the provisions of 40 CFR Part 63, Subpart LLL is exempted from any otherwise applicable new source performance standard contained in 40 CFR Part 60, Subpart F. (63.1356)

22. Opacity Limits

These limits apply only to those sources which are referred to this Condition throughout this permit. **This condition applies to all activities at the plant except the front end loaders and the kiln and cooler.**

22.1 Opacity shall not exceed 20%, except as provided for in Condition 22.2, below (Colorado Regulation No. 1, II.A.1).

At the time of inspection, no opacity was noted.

22.2 Opacity of emissions during fire building, cleaning of fire boxes, soot blowing, start-up, process modifications or adjustment of control equipment shall not exceed 30% for a period or periods aggregating more than six (6) consecutive minutes in any sixty (60) consecutive minutes (Colorado Regulation No. 1, Section II.A.4).

Opacity of emissions during fire building, cleaning of fire boxes, soot blowing, start-up, process modifications or adjustment of control equipment shall not exceed 30%.

Compliance with these limits shall be monitored by meeting the requirements of Condition No. 16 of this permit, and by meeting the following Bagfilter Operation and Maintenance requirements.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or the permittee's operating experience. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

Routine maintenance and operational procedures performed on the baghouse is conducted in accordance with the manufacturer's specifications. A monthly work order is produced for with a checklist for all the baghouses. They are inspected for problems and bags are replaced if needed. These work orders are kept on site for review upon request. Records of emissions calculations are maintained and calculated using the test results and the operating hours.

23. PM Stack Testing

This requirement applies only to those sources which are referred to this condition throughout this permit.

This condition specifically refers to the kiln, cooler, raw material dryer, finish mill system (points 009, 010, 010A, 011, 012, and 013) CKD and benification dust to the silo (P040A & B).

Performance testing for particulate emissions shall be performed within one year of issuance of this permit, and every year thereafter, in accordance with the requirements and procedures set forth in EPA Test Method 5 as set forth in 40 CFR Part 60, Appendix A. A stack testing protocol shall be submitted for Division approval at least thirty (30) calendar days prior to performance of the test. No stack test shall be performed without prior written approval by the Division. The Division reserves the right to

witness the test. The required number of copies of the compliance test results shall be submitted to the Division within forty-five (45) calendar days of the completion of the test.

Performance testing has been completed for this permit term. Stack testing was done in September 1999 for the finish mill, in January for the CKD points, in February for the kiln and in April for the clinker cooler and raw materials dryer. Results have been submitted to the Division for review and copies of some of the results are attached to this report. Southdown did have a problem with the initial stacktest of the raw materials dryer. Particulate emissions were 53.6 tons/year and the permit allows for 22.8 tons/year. The dryer was shut down and maintenance was performed and 40 bags in the baghouse were replaced. A second stack test done in June showed they were in compliance with their permit limit, as indicated in this report. Also as a result of this, they will be installing a broken bag detector and perform quarterly fluorescent dye tests to check for problems. This will be incorporated into their Bagfilter Operating and Maintenance Plan.

24. National Emissions Standards for Hazardous Air Pollutants From the Portland Cement Manufacturing Industry

This source will be subject to this MACT in June of 2002. They have submitted their initial notification to the state (due 10/12/99) and are working on meeting the other requirements. They did some diagnostic testing in February and March of 2000 already for dioxins and flourans from the kiln stack. CDPHE also did stack testing in May of the this year for dioxins/flourans. These limits were below the NESHAP limit of 0.2 ng/dscm. The average of the three runs by APT were 0.2 ng/dscm, CDPHE's results were .004 ng/dscm. See attached copy of results.

SOURCE COMPLIANCE:

In compliance with permit conditions, except the following:

- **P002 Raw Materials Drying (Condition 5)** - The dryer heat input mmBTU was exceeded in 1999 due to an inaccurate gas meter. A letter explaining the problem was sent to APCD and the number used for the heat content has been corrected so this should not be a problem for the 2000 calculations.
- **P002 Raw Materials Drying (Condition 5)** - PM_{10} lbs/hour limit was exceeded by 0.5 lbs/hour. The total for the year is under permit limits, but the lbs/hour was exceeded since the source operated less than 7,000 hours. The results of a stack test done in June 2000 showed they were in compliance with this limit. Results were 2.3 lbs/hour of particulates.
- **Opacity - Daily Plant Walkthrough (Condition 16)** - Records are maintained although not all the records were clear about what actions were taken and when the problem was corrected. There is space on the bottom for this, so Steve said he make sure they were better about documenting corrections.
- **PM Stack Testing (Condition 23)** - The initial stack test on the raw materials dryer failed. The particulate emissions were more than twice what they were permitted for (53.6 tons/year vs. 22.8 tons/year). The dryer was shut down for maintenance, bags in the baghouse were replaced and at the retest they were in compliance. A Compliance Order on Consent was issued in July and finalized in September 2000 agreeing to a fine and SEP.